Created 2010/9 mid term ….
ASSALAM O ALIKUM all fellows
ALL IN ONE MTH301 MIDTERM PAPERS
solved (4)
SOLVED BY Farhan & Ali
BS (cs) 2nd sem
Hackers Group
Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

MIDTERM EXAMINATION

Spring 2010
MTH301- Calculus II

Ref No: 1499814
Time: 60 min
Marks: 40

Student Info	
Student ID:	
Center:	
Exam Date:	

For Teacher's Use Only									
Q No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Total**
Marks									
Q No.	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
Marks									
Q No.	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	
Marks									
Q No.	**25**	**26**							

Marks									

Question No: 1 (Marks: 1) - Please choose one

Every point in three dimensional space can be described by ------- coordinates.

► Two

► **Three**

► Four

► Eight

Question No: 2 (Marks: 1) - Please choose one

What are the direction cosines for the line joining the points (1, 3, 2) and (7, -2, 3)?

► $\frac{1}{7}, \frac{-3}{2}$ *and* $\frac{2}{-3}$

► $\frac{7}{11}, \frac{-6}{11}$ *and* $\frac{6}{11}$

► $\frac{8}{3\sqrt{10}}, \frac{1}{3\sqrt{10}}$ *and* $\frac{5}{3\sqrt{10}}$

► $\frac{6}{\sqrt{62}}, \frac{-5}{\sqrt{62}}$ *and* $\frac{1}{\sqrt{62}}$

Question No: 3 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ----------

► Direction cosines

► Direction ratios

► **Direction angles**

Question No: 4 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

► A point on y-axis (Farhan)

► Plane parallel to xy-plane

► Plane parallel to yz-axis

► **Plane parallel to xz-plane (Ali)**

Question No: 5 (Marks: 1) - Please choose one

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is

► **Entire 3D-Space**

► Entire 3D-Space except origin

► $(0,\infty)$

► $(-\infty, \infty)$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = xy^2 - y^3 + \ln x$

then $\frac{\partial 2 f}{\partial x_2} =$

- $2xy + \frac{1}{x_2}$
- $2y + \frac{1}{x_2}$
- $2xy - \frac{1}{x_2}$
- $2y - \frac{1}{x_2}$

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y$ *where* $x^2 = +3t\ 1$ *and* $y = 2t$. Which one of the following is true?

- $\frac{df}{dt} = - + 4t\ \ 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt}\ \ df^{\ 2}\ \ 8t\ \ 1$

► $=-10t + +$

dt

Question No: 8 (Marks: 1) - Please choose one

For a function $f\ x\ y\ z(, ,)$, the equation $\partial\partial 22xf + \partial\partial 22yf + \partial\partial 22zf = 0$ is known as ----------

► Gauss Equation

► Euler's equation

► Laplace's Equation (Ali)

► Stoke's Equation

Question No: 9 (Marks: 1) - Please choose one

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

► 4.5

► 6.2

► 5.1

► 4.2

Question No: 10 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x + y^2}$$

- $f(x, y)$ is continuous at origin
- $\lim_{(x, y) \to (0, 0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x, y) \to (0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 11 (Marks: 1) - Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R.??(ali)
- **Radius of inscribed circle in R**
- Distance between two endpoints of R.
- None of these

Question No: 12 (Marks: 1) - Please choose one

Which of the following formula can be used to find the Volume of a parallelepipedr r r with adjacent edges formed by the vectors *a b and c*, ? r r r *a* × ×(*b c*)

► |r r r *a* g g(*b c*)|

► | r r r *a* g(*b* × *c*|

► |r r r *a* × (*b* g*c*)|

►

Question No: 13 (Marks: 1) - Please choose one

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

► Parallel

► **Perpendicular**

► In opposite direction

Question No: 14 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ---------- ► perpendicular

► parallel

► **In opposite direction (Farhan)**

Question No: 15 (Marks: 1) - Please choose one

Let the function *f x y*(,) has continuous second-order partial derivatives

(f_{xx} , f_{yy} *and* f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}2\,(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► **Relative maximum at** (x_0, y_0)

► Relative minimum at (x_0, y_0)
► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 16 (Marks: 1) - Please choose one

Let the function $f\,x\,y(\ ,\ \ )$ has continuous second-order partial derivatives

(f_{xx} , f_{yy} *and* f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}2(x_0, y_0)$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 17 (Marks: 1) - Please choose one

If $R = \{(x\,y,\,)/0 \le \le x \quad 2$ *and* $1 \le \le y \quad 4\}$, *then*
$\iint (6x^2 + 4xy\ dA^3) \quad =$

R

$\int_{1}^{4}\int_{0}^{2}(6x^2 + 4xy\ dydx^3)$

$\int_{0}^{2}\int_{1}^{4}(6x^2 + 4xy\ dxdy^3)$

$\int_{0}^{4}\int_{1}^{2}(6x^2 + 4xy\ dxdy^3)$

$\int_{1}^{4}\int_{2}^{1}(6x^2 + 4xy\ dxdy^3)$

Question No: 18 (Marks: 1) - Please choose one

If $R = \{(x\ y,\ )/0 \leq \leq x \quad 2 \text{ and } -1 \leq \leq y\ 1\}$, *then*

$\iint_{R}(x + 2y\ dA^2) \quad =$

$\int_{1}^{2}\int_{-1}^{0}(x + 2y\ dydx^2)$

$\int_{2}^{-1}\int_{0}^{1}(x + 2y\ dxdy^2)$

$\int_{1}^{2}\int_{-1}^{0}(x + 2y\ dxdy^2)$

$$\int_{2}^{} \int_{0}^{} (x+ 2y\, dxdy^2)$$
$$\int_{1}^{} \int_{-1}^{}$$

Question No: 19 (Marks: 1) - Please choose one

If $R = \{(x\, y,)/2 \leq \leq x \quad 4 \textit{and } 0 \leq \leq y\ 1\}, \textit{then}$
$\iint (4xe^{2y})dA =$
R

- $\int_{0}^{1} \int_{2}^{4} (4 \quad {}^{2y}) \, xedydx$
- $\int_{0}^{1} \int_{2}^{4} (4 \quad {}^{2y}) \, xedxdy$
- $\int_{1}^{4} \int_{0}^{2} (4 \quad {}^{2y}) \, xedxdy$
- $\int_{1}^{4} \int_{0}^{2} (4 \quad {}^{2y}) \, xedydx$

Question No: 20 (Marks: 1) - Please choose one

If $R = \{(x\, y,)/0 \leq \leq x \quad 4 \textit{and } 0 \leq \leq y\ 9\}, \textit{then}$
$\iint (3x-4x\, xy\sqrt{dA}) \quad =$
R

$$\int_{0}^{9} \int_{0}^{4} (3x-4x\, xy\sqrt{dydx})$$

4 9

► $\int\int (3x-4x\ xy\ d\sqrt{x}dy)$

0 4

9 0

► $\int\int (3x-4x\ xy\ d\sqrt{x}dy)$

4 0

4 9

► $\int\int (3x-4x\ xy\ d\sqrt{y}dx)$

0 0

Question No: 21 (Marks: 2)

Suppose that the surface $f\ x\ y\ z(\ ,\ ,\ )$ has continuous partial derivatives at the point $(a\ b\ c,\ ,\ )$. Write down the equation of tangent plane at this point.

Question No: 22 (Marks: 2)

Evaluate the following double integral.

$\iint(\ 3x-\ y\ dy\ dx)$

$\iint(\ 3x\ y^2\ \ /\ 2-\ xy^2\ /\ 2)\ dx\iiint(\ 3x-\ y\ dy\ dx)$

$\int\frac{3x^1y}{1}-\frac{y^2}{2}\,dx$

$\frac{3x^2y}{2}-\frac{xy^2}{2}+c$

Answer

$\int\frac{3x^1y}{1}-\frac{y^2}{2}\,dx$

$\frac{3x^2y}{2}-\frac{xy^2}{2}+c$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$\iint(3+ 2x-3y^2)\, dx\, dy$

$=\int(3x+ x^2 +3xy^2)\, dy$

$= 3xy + x\, y^2 +3xy^3$

$\iint(3+ 2x-3y^2)\, dx\, dy$

$=\int(3x+ x^2 +3xy^2)\, dy$

$= 3xy + x\, y^2 +3xy^3$

Question No: 24 (Marks: 3)

Let $f\, x\, y\, z(, ,) = yz^3 -2x^2$ *Find the gradient of* f .

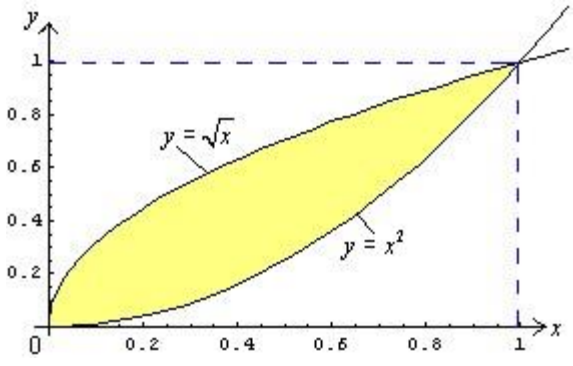
Question No: 25 (Marks: 5)

Find all critical points of the function

$f\ x\ y(,) = y^2 + xy+3y + 2x+3$

Question No: 26 (Marks: 5)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



PAPER NO.2

1. Every real number corresponds to_____on the co-ordinate line.

- Infinite number of points
- Two points (one positive and one negative)
- A unique point??f
- None of these

2. There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

3. Which of the following is associated to each point of three dimensional spaces?

- A real number
- An ordered pair
- An ordered triple
- **A natural Number**

4. All axes are positive in _____octant.

- **First**
- Second
- Fourth
- Eighth

5. The spherical co-ordinates of a point are $(\sqrt{3}, \pi, \frac{\pi}{2})$.What are its cylindrical co-ordinates?

- $(\sqrt{3}, \frac{\pi}{3}, 3)$
- $(2, \frac{\pi}{2}, 2)$

□√3cosπ, √3sinπ,0□□

➢ □□ 3 3 □

□√3sinππ, , √3cosπ□□

➢ □□ 3 2 3□

□√3,π,0□□

➢ □□ 3 □

6. Suppose $f(x, y) = -xy\ 2y$ *where* $x^2 = +3t\ 1$ *and* $y = 2t$. Which one of the following is true?

➢ $\frac{df}{dt} = -\ +4t \quad 2$

➢ $\frac{df}{dt} = -16t\ -t$

➢ $\frac{df}{dt} = 18t + 2$

➢ $\frac{df}{dt} = -10t + +8t^2 \quad 1$

7. Let $w = f(x, y, z)$ *and* $x = g(r, s), \ y = h(r, s), \ z = t(r, s)$ then by chain rule $\frac{\partial w}{\partial r} =$

□ $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

□ $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$

➢ $\frac{\partial}{\partial x}\frac{\partial}{\partial r}\frac{\partial w}{\partial x}x\,x + \frac{\partial}{\partial y}\frac{\partial}{\partial r}\frac{\partial w}{\partial y}y\,y + \frac{\partial}{\partial z}\frac{\partial}{\partial r}\frac{\partial w}{\partial z}z\,z$

□ $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3and angle between them when placed tail to tail is 45 degrees. What is $\vec{a}.\vec{b}$?

- 4.5
- **6.2** □□5.1

□□**4.2**

9. Is the function $f(x,y)$ continuous at origin? If not, why?

$$f(x,y) = \begin{cases} 0 & if \ x \geq 0 \ and \ y \geq 0 \\ 1 & otherwise \end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

10. Is the function $f(x,y)$ continuous at origin? If not, Why?

$$f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & if (x,y) \neq 0 \\ 0 & if \ (x,y) = 0 \end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ **is not defined**
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R
- **Radius of inscribed circle in R.**
- Distance between two endpoints of R.
- None of these

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

- $|\vec{a} \times (\vec{b} \times \vec{c})|$
- $|\vec{a} . (\vec{b} . \vec{c})|$
- $|\vec{a} . (\vec{b} \times \vec{c})|$
- $|\vec{a} \times (\vec{b} . \vec{c})|$

13. Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- Parallel
- **Perpendicular**
- In opposite direction
- Same direction

14. By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both _____ on R.

Absolute maximum and absolute minimum value

- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

15. Let the function $f(x, y)$ has continuous second-order partial derivatives (f_{xx}, f_{yy} *and* f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if $D > 0$ *and* $f_{xx}(x_0, y_0) < 0$ *then f has* _____ .

Relative maximum at (x_0, y_0)

- Relative minimum at (x_0, y_0)
- Saddle point at (x_0, y_0)
- No conclusion can be drawn.

16. Let the function $f(x, y)$ has continuous second-order partial derivatives

(f_{xx} , f_{yy} and f_{xy}) in some circle centered at a critical point (x_0, y_0) and let
$D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if
if $D = 0$ then _____.

- f has relative maximum at (x_0, y_0)
- f has relative minimum at (x_0, y_0)
- f has saddle point at (x_0, y_0)
- **No conclusion can be drawn.**

17. *If* $R = R_1 \cup R_2$, *where* R_1 *and* R_2 are no over lapping regions then
$\iint_{R1} f(x, y)\, dA + \iint_{R2} f(x, y)\, dA =$

- $\iint_R f(x, y)\, dA$
- $\iint_{R1} f(x, y)\, dA \cup \iint_{R2} f(x, y)\, dA$
- $\iiint_R f(x, y)\, dV$
- $\iiint_R f(x, y)\, dV \cap \iint_{R2} f(x, y)\, dA$

18. *If* $R = \{(x, y) / 0 \le x \le 2$ *and* $1 \le y \le 4\}$, *then* $\iint_R (6x^2 + 4xy\, dA^3) =$

- $\int_1^4 \int_0^2 (6x^2 + 4xy)\, dydx$
- $\int_0^2 \int_1^4 (6x^2 + 4xy)\, dydx$
- $\int_1^4 \int_0^2 (6x^2 + 4xy)\, dydx$

- $\int_{2}^{4}\int_{0}^{1}\left(6\ ^{2}+4\ ^{3}\right)x\ \ xy\ \ dydx$

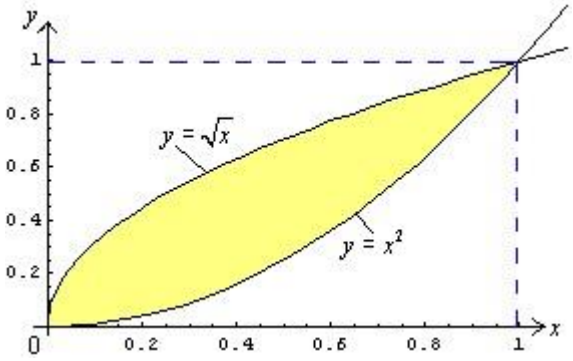
19. *If* $R=\{(x\ y,\ )/2\leq\leq x\ \ 4$ *and* $0\leq\leq y\ 1\}$, *then* $\iint(4xe^{2y})dA=$
R

- $\int_{0}^{1}\int_{2}^{4}\left(4\ ^{2y}\right)xe\ \ dydx$
- $\int_{0}^{1}\int_{2}^{4}\left(4\ ^{2y}\right)xe\ \ dxdy$
- $\int_{1}^{4}\int_{0}^{2}\left(4\ ^{2y}\right)xe\ \ dxdy$
- $\int_{1}^{4}\int_{0}^{2}\left(4\ ^{2y}\right)xe\ \ dydx$

20. *If* $R=\{(x\ y,\ )/0\leq\leq x\ \ 4$ *and* $0\leq\leq y\ \ 9\}$ $\sqrt{x}$ *then* $\iint(3x-4x\ xy\ dA)$
$=$
R

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\ x\sqrt{y}\,dydx)$
- $\int_{0}^{9}\int_{0}^{4}(3x-4x\ x\sqrt{y}\,dxdy)$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\ x\sqrt{y}\,dxdy)$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\ x\sqrt{y}\,dydx)$

21. Suppose that the surface $f\,x\,y\,z(\,,\,,\,)$ has continuous partial derivatives at the point $(\,a\,b\,c,\,,\,)$ Write down the equation of tangent plane at this point.
22. Evaluate the following double integral $\iint(12xy^2 - 8x^3)\,dydx$.
23. Evaluate the following double integral $\iint(\,3+\,2x - 3y^2\,)\,dxdy$
24. Let $f\,x\,y\,z(\,,\,,\,) = xy\,e^{2\,z}$ Find the gradient of f .
25. Find, Equation of Tangent plane to the surface $f\,x\,y\,z(\,,\,,\,) = x^2 + y^2 + -z\,9$ at the point (1,2,4) .
26. Use the double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2 and y = \sqrt{x}$.



MIDTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 3)

Time: 60 min
Marks: 40

Student Info	
StudentID:	
Center:	**OPKST**
ExamDate:	**6/3/2010 12:00:00 AM**

For Teacher's Use On y

Q No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Total**
Marks									

Q No.	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
Marks									

Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26							
Marks									

MTH301 Calculus II

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Every real number corresponds to ---------- on the co-ordinate line.

Answer (Please select your correct option)

- Infinite number of points
- Two points (one positive and one negative)
- A unique point
- None of these

MTH301 Calculus II

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

MTH301 Calculus II

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point of three dimensional space?

Answer (Please select your correct option)

- A real number
- An ordered pair
- An ordered triple
- A natural number

MTH301 Calculus II

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

All axes are positive in ---------- octant.

Answer (Please select your correct option)

- First
- Second
- Fourth
- Eighth

MTH301 Calculus II

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylinderical co-ordinates?

Answer (Please select your correct option)

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

MTH301 Calculus II

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

Answer (Please select your correct option)

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt} = -10t^2 + 8t + 1$

MTH301 Calculus II

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r}$ =

Answer (Please select your correct option)

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

MTH301 Calculus II

Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

Answer (Please select your correct option)

- 4.5
- 6.2
- 5.1
- 4.2

MTH301 Calculus II

Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & \text{If } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{Otherwise} \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 10 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \dfrac{3x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Answer (Please select your correct option)

- Area of R.
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

MTH301 Calculus II

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\bar{a}$, $\bar{b}$ *and* $\bar{c}$?

Answer (Please select your correct option)

- $\left|\bar{a} \times (\bar{b} \times \bar{c})\right|$
- $\left|\bar{a} \bullet (\bar{b} \bullet \bar{c})\right|$
- $\left|\bar{a} \bullet (\bar{b} \times \bar{c})\right|$
- $\left|\bar{a} \times (\bar{b} \bullet \bar{c})\right|$

MTH301 Calculus II

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

Answer (Please select your correct option)

- Parallel
- Perpendicular
- In opposite direction

MTH301 Calculus II

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

Answer (Please select your correct option)

- Absolute maximum and absolute minimum value
- Relative maximum and relative minimum value

MTH301 Calculus II

Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Answer (Please select your correct option)

- (●) Relative maximum at (x_0, y_0)
- ( ) Relative minimum at (x_0, y_0)
- ( ) Saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D = 0$ then ----------

Answer (Please select your correct option)

- (●) f has relative maximum at (x_0, y_0)
- ( ) f has relative minimum at (x_0, y_0)
- ( ) f has saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

Answer (Please select your correct option)

- $\iint_{R} f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iiint_{R} f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

MTH301 Calculus II

Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$ *,then*

$$\iint_{R} (6x^2 + 4xy^3)dA =$$

Answer (Please select your correct option)

- $\int_{1}^{4}\int_{0}^{2} (6x^2 + 4xy^3)dydx$
- $\int_{0}^{2}\int_{1}^{4} (6x^2 + 4xy^3)dxdy$
- $\int_{1}^{4}\int_{0}^{2} (6x^2 + 4xy^3)dxdy$
- $\int_{2}^{4}\int_{0}^{1} (6x^2 + 4xy^3)dxdy$

MTH301 Calculus II

Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, then

$\iint_R (4xe^{2y})dA =$

Answer (Please select your correct option)

- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$ ✓
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

MTH301 Calculus II

Question No : 20 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 0 \le x \le 4 \text{ and } 0 \le y \le 9\}$, then

$\iint_R (3x - 4x\sqrt{xy})dA =$

Answer (Please select your correct option)

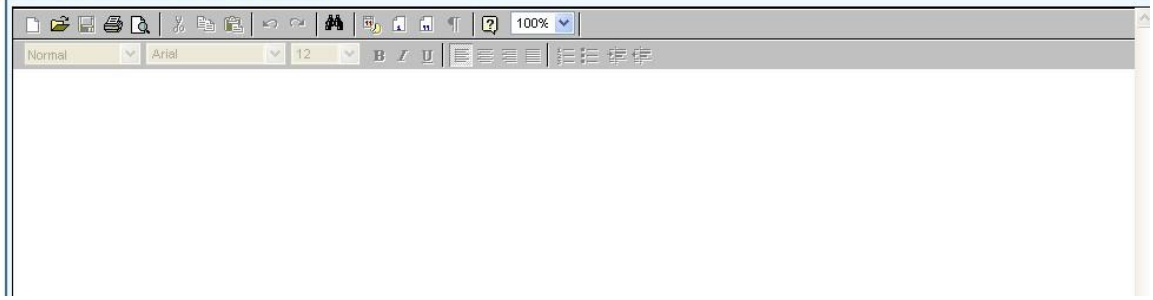
- $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$
- $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$
- $\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$
- $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$ ✓

MTH301 Calculus II

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c). Write down the equation of tangent plane at this point.

Answer (Please click here to Add Answer)



MTH301 Calculus II

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following double integral.

$$\iint (12xy^2 - 8x^3)\, dy\, dx$$

MTH301 Calculus II

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Evaluate the following double integral.

$$\iint (3 + 2x - 3y^2)\, dx\, dy$$

Answer (Please click here to Edit Answer)

MTH301 Calculus II

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Let $f(x, y, z) = xy^2e^z$

Find the gradient of f.

Answer (Please click here to Add Answer)

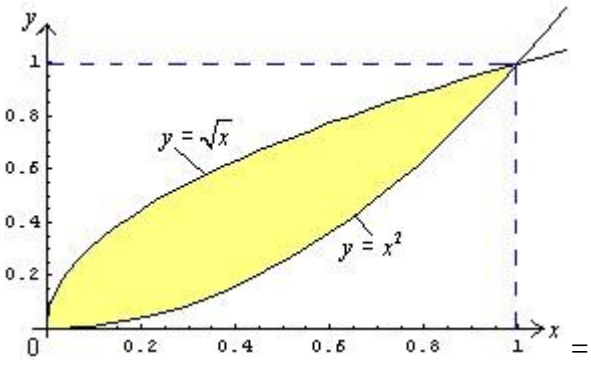
MTH301 Calculus II

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find, Equation of Tangent plane to the surface $f(x, y, z) = x^2 + y^2 + z - 9$ at the point $(1, 2, 4)$

Answer (Please click here to Add Answer)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.

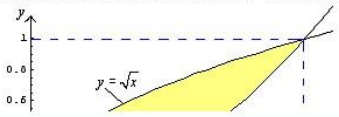


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MTH301 Calculus II

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



Answer (Please click here to Add Answer)

SOLVED BY Farhan & Ali
BS (cs) 2nd sem
Hackers Group
Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

ASSALAM O ALAIKUM

All Dear fellows

ALL IN ONE MTH301 Calculus II

Midterm solved papers

Created BY Ali Shah

From Sahiwal

BSCS 4th semester alaoudin.bukhari@gmail.com

Remember me in your prayers

MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 3)

1. Every real number corresponds to_____on the co-ordinate line.

- **Infinite number of points**
- Two points (one positive and one negative)
- A unique point
- None of these

2. There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of real numbers
- Set of integers
- **Set of natural numbers**
- Set of rational numbers

3. Which of the following is associated to each point of three dimensional space?

- A real number
- An ordered pair
- An ordered triple
- **A natural Number**

4. All axes are positive in _____octant.

- **First**
- Second
- Fourth
- Eighth

5. The spherical co-ordinates of a point are $\left(3, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylindrical co-ordinates?

$$\left(\frac{3}{2}, \frac{\pi}{2}, 0\right)$$

$$\left(\frac{3\sqrt{3}}{2}, \frac{\pi}{2}, 0\right)$$

- $\left(3\cos\frac{\pi}{3}, 3\sin\frac{\pi}{3}, 0\right)$

$$\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}, 0\right)$$

- $\left(3\sin\frac{\pi}{3}, \frac{\pi}{3}, 3\cos\frac{\pi}{3}\right)$

$$\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}, 3\right)$$

- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

$$\left(3, \frac{\pi}{3}, 0\right)$$

6. Suppose $f(x, y) = -xy^2 + 2y$ where $x = t^2 + 3t + 1$ and $y = t$. Which one of the following is true?

- $\frac{df}{dt} = -4t^2 + 2$

$$\frac{df}{dt}$$

- $\frac{df}{dt} = -16t - t$

$$\frac{df}{dt}$$

- $\frac{df}{dt} = 18t + 2$

$$\frac{df}{dt}$$

- $\frac{df}{dt} = -10t^2 + 8t + 1$

7. Let $w = f\,x\,y\,z$ *and* $x(\ ,\ ,\ )$ $= g\ r\ s(\quad)$, $y,\ = h\ r\ s(\quad)$,$z =,\ \ t\ r\ s(\quad)$ then by chain rule $\frac{\partial w}{\partial r} =$

- $\frac{\partial w}{\partial x}\frac{x\partial}{\partial r} + \frac{\partial w}{\partial y}\frac{y\partial}{\partial r} + \frac{\partial w}{\partial z}\frac{z\partial}{\partial r}$
- $\frac{\partial}{\partial r}\frac{\partial w}{\partial x}\frac{x}{\partial} + \frac{\partial}{\partial r}\frac{\partial w}{\partial y}\frac{y}{\partial} + \frac{\partial}{\partial r}\frac{\partial w}{\partial z}\frac{z}{\partial}$
- $\frac{\partial w}{\partial x}\frac{x\partial}{\partial} + \frac{\partial w}{\partial y}\frac{y\partial}{\partial} + \frac{\partial w}{\partial z}\frac{z\partial}{\partial}$

$\frac{\partial}{\partial x}\frac{\partial r}{\partial}\frac{r}{s} \quad \frac{\partial}{\partial y}\frac{\partial r}{\partial}\frac{r}{s} \quad \frac{\partial}{\partial z}\frac{\partial r}{\partial}\frac{r}{s}$

$\frac{\partial w}{\partial r}\frac{\partial r}{\partial} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial}$

$\frac{\partial r}{\partial x} \quad \frac{\partial r}{\partial y} \quad \frac{\partial r}{\partial z}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3and angle between them when placed tail to tail is 45 degrees. What is$\vec{a}.\vec{b}$?

- 4.5
- **6.2**
- 5.1
- 4.2

9. Is the function $f(x\,y,)$ continuous at origin? If not, why? $f(x,y) = \begin{cases} 0 & if\ \ x \geq 0\ and\ y \geq 0 \\ 1 & otherwise \end{cases}$

- $f(x\,y,)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $(x\,y,\ \lim)\rightarrow(0,0)\ f\,x\,y(\ ,\ )$ does not exist
- $f(0,{}^{0})$ is defined and $(x\,y,\ {}^{\lim})_{\rightarrow}(0,0_{)}\ f\,x\,y(\ ,\ )$ exist but these two numbers are not equal.

10. Is the function $f\,x\,y(\ ,\ )$continuous at origin? If not, Why? $f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & if(x,y) \neq 0 \\ 0 & if\ (x,y) = 0 \end{cases}$

- $f\,x\,y(\ ,\ )$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $(x\,y,\ \lim)\rightarrow(0,0)\ f\,x\,y(\ ,\ )$ does not exist

- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R
- Radius of inscribed circle in R.
- Distance between two endpoints of R. None of these

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

- $|\vec{a} \times (\vec{b} \times \vec{c})|$
- $|\vec{a} . (\vec{b} . \vec{c})|$
- $|\vec{a} . (\vec{b} \times \vec{c})|$
- $|\vec{a} \times (\vec{b} . \vec{c})|$

13. Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- Parallel
- Perpendicular
- In opposite direction
- Same direction

14. By Extreme Value Theorem, if a function $f(x,y)$ is continuous on a closed and bounded set R, then $f(x,y)$ has both _____ on R.

- Absolute maximum and absolute minimum value
- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

15. Let the function $f(x,y)$ has continuous second-order partial derivatives (f_{xx} , f_{yy} *and* f_{xy}) in some circle centered at a critical point $(x_0, y_0)$and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$if $D > 0$ *and* $f_{xx}(x_0, y_0) < 0$ *then f has* _____ .

- Relative maximum at $(x\ y_0, {}_0)$
- Relative minimum at $(x\ y_0, {}_0)$
- Saddle point at $(x\ y_0, {}_0)$
- No conclusion can be drawn.

16. Let the function $^{f}(x\ y)$ has continuous second-order partial derivatives (f_{xx} , f_{yy} *and* f_{xy}) in some circle centered at a critical point $(x\ y_0, {}_0)$and let $D = f_{xx}(x\ y_0, {}_0)\ f_{yy}(x\ y_0, {}_0) - f_{xy}^2(x\ y_0, {}_0)$if

if D=0 then _____.

- f has relative maximum at $(x\ y_0, {}_0)$
- f has relative minimum at $(x\ y_0, {}_0)$
- f has saddle point at $(x\ y_0, {}_0)$ No conclusion can be drawn.

17. *If* $R = R_1 \cup R_2$, *where* R_1 *and* R_2 are no over lapping regions then

$\iint_{R1} f(x\ y)\,dA(\ ,\ ) + \iint_{R2} f(x\ y)\,dA(\ ,\ ) =$

- $\iint_{R} f(x\ y)\,dA(\ ,\ )$
- $\iint_{R1} f(x\ y)\,dA(\ ,\ ) \cup \iint_{R2} f(x\ y)\,dA(\ ,\ )$
- $\iint_{R} f(x\ y)\,dV(\ ,\ )$
- $\iint_{R} f(x\ y)\,dV(\ ,\ ) \cap \iint_{R_2} f(x\ y)\,dA(\ ,\ )$

18. *If* $R = \{(x\ y,) / 0 \le \le x \quad 2 \text{ and } \le \le 1\ y\}$ *then* $4\iint_R (6x^2 + 4xy\,dA^3) =$

- $\iint (6x^2 + 4xy^3)dydx$

- $\int_{1}^{4}\int_{0}^{2}\left(6^{2}+4^{3}\right)$ x xy $dydx$
- $\int_{1}^{4}\int_{0}^{2}\left(6^{2}+4^{3}\right)$ x xy $dydx$
- $\int_{2}^{4}\int_{0}^{1}\left(6^{2}+4^{3}\right)$ x xy $dydx$

19. *If* $R=\{(x,y)/2\le x\le 4 \text{ and } 0\le y\le \}$

- $\int_{0}^{2}\int_{1}^{4}\left(4^{2y}\right)$ xe $dydx$
- $\int_{0}^{2}\int_{1}^{4}\left(4xe^{2y}\right)dxdy$ *then* 1, $\iint_R(4xe^{2y})dA=$
- xe $dxdy$
- xe $\int_{1}^{4}\left(4^{2y}\right)$ $dydx$
- xe $\int_{1}^{4}\left(4^{2y}\right)$ $dydx$

20. *If* $R=\{(x,y)/0\le x\le 4 \text{ and } \le y\le 0\}$ *then* $9\iint_R(3x-4x\,xy\,dA)$ $=\sqrt{}$

- $\int_{0}^{9}\int_{0}^{4}\left(3x-4xx\sqrt{y}dydx\right)$
- $\int_{0}^{9}\int_{0}^{4}\left(3x-4xx\sqrt{y}dxdy\right)$
- $\int_{0}^{9}\int_{0}^{4}\left(3x-4xx\sqrt{y}dxdy\right)$
- $\int_{0}^{9}\int_{0}^{4}\left(3x-4x\sqrt{xy}\,dydx\right)$

21. Suppose that the surface $f\,x\,y\,z(\,,\,,\,)$ has continuous partial derivatives at the point $(a\ b\ c,\,,\,)$Write down the equation of tangent plane at this point.

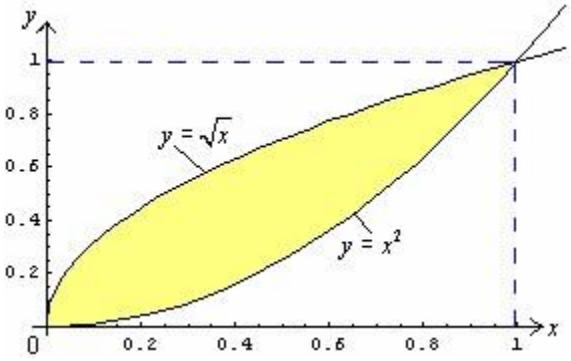
22. Evaluate the following double integral $\iint(12xy^2 - 8x^3)dydx$.

23. Evaluate the following double integral $\iint(3+ -2x \quad 3y^2)dxdy$

24. Let $f\,x\,y\,z(\,,\,,\,)= xy\,e^{2\,z}$ Find the gradient of f .

25. Find, Equation of Tangent plane to the surface $f\,x\,y\,z(\,,\,,\,)= + + -x^2 \quad y^2 \quad z \quad$ 9 at the point(1,2,4).

26. Use the double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x$ *and* $y^2 \quad = x$.$\sqrt{}$



MIDTERM EXAMINATION

Fall 2009

MTH301- Calculus II

Question No: 1 (Marks: 1)- Please choose one

i. _____ Let x

be any point on co-ordinate line. What does the inequality -3 < x < 1 means?

a. ► The set of all integers between -3 and 1

b. ► The set of all natural numbers between -3 and 1.

c. ► The set of all rational numbers between -3 and 1

d. **► The set of all real numbers between -3 and 1**

Question No: 2 (Marks: 1)- Please choose one

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► A real number

► A rational number

► A natural number

Question No: 3 (Marks: 1)- Please choose one

Which of the following set is the union of set of all rational and irrational numbers?

► Set of rational numbers

► Set of integers

► Set of

real numbers

► Empty

set.

Question No: 4 (Marks: 1)- Please choose one

---

is an example of ----------

► **Irrational numbers**

► Rational numbers

► Integers

► Natural numbers

Question No: 5 (Marks: 1)- Please choose one

Which of the following is associated to each point on a plane?

► A real number

► A natural number

► An ordered pair

► An ordered triple

Question No: 6 (Marks: 1)- Please choose one

---------- planes intersect at right angle to form three dimensional space.

► **Three**

► Four

► Eight

► Twelve

Question No: 7 (Marks: 1)- Please choose one

At each point of domain, the function ----------

► **Is defined**

► Is continuous

► Is infinite

► Has a limit

Question No: 8 (Marks: 1)- Please choose one

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

► $y = ax^2 + b \quad (a \neq 0)$
► $x = ay^2 + b \quad (a \neq 0)$

► $y = ax^2 + bx + c \quad (a \neq 0)$

► $x = ay^2 + by + c \quad (a \neq 0)$

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Question No: 9 (Marks: 1)- Please choose one

The

□ □ □ □ **4** □

□2, , 0□

spherical co-ordinates (□ □ □, ,), of a point are . What are the rectangular co-ordinates of this point?

- $(0, 0, \sqrt{2})$ ►
- (2, 0, 0) ►
- (0, 0, 2) ►
- $(\sqrt{2}, 0, \sqrt{2})$ ►

Question No: 10 (Marks: 1)- Please choose one

Let

$f(x, y) = y\, x\, e^{24x} + 2$

.

$$\frac{\partial^5 f}{\partial y^3 \partial x^2}$$

Which method is best suited for evaluation of ?

- ► Normal method of finding the higher order mixed partial derivatives
- ► Chain Rule
- ► Laplacian Method
- **► Euler's method for mixed partial derivative**

Question No: 11 (Marks: 1)- Please choose one

Suppose $f(x, y) = x e^{3xy}$. Which one of the following is correct?

► $\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

► $\frac{\partial f}{\partial x} = 3x e^{2xy} + x e^{4xy}$

$\frac{\partial f}{\partial x} = 3 x e^{x y}$

► ∂x

► $\frac{\partial f}{\partial x} = 3x^2 y e^{xy}$

Question No: 12 (Marks: 1)- Please choose one

Suppose $f(x, y) = x e^{3xy}$. Which one of the statements is correct?

► $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

► $\frac{\partial f}{\partial y} = x^3 exy$

$\frac{\partial f}{\partial y} = x^4 e^{xy}$

► $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 13 (Marks: 1)- Please choose one

Suppose $f(x, y) = 2xy$ *where* $x = t^2 + 1$ *and* $y = 3 - t$. Which one of the following is true?

$\frac{df}{dt} = -6t$

► $\frac{df}{dt} = -6t^2$

$\frac{df}{dt} = 4t^3 +$

-6
t
6
dt

▶

$\frac{df}{dt} = -\ +6t\ 12t\ -2$

▶

Question No: 14 (Marks: 1)- Please choose one

Let

i, j and k be unit vectors in the direction of x-axis, y-axis and z-axis respectively. Suppose that

$\rightarrow$

$a = 2i + 5j - k$ $\vec{a}$. What is the magnitude of vector ?http://vustudents.ning.com

▶ 6

▶ 30

▶ $\sqrt{30}$

$\sqrt{28}$ ▶

Question No: 15 (Marks: 1)- Please choose one

Is $f\ x\ y($,) the function continuous at origin? If not, why?

xy
$f\ x$
y

(,) =

$$\frac{x^2 + y^2}{}$$

$f(x, y)$

► is continuous at origin

$\lim_{(x, y) \to (0, 0)} f(x, y)$

► **does not exist**

$f(0, 0)$ $\lim_{(x, y) \to (0, 0)} f(x, y)$

► is defined and exists but these two numbers are not equal.

Question No: 16 (Marks: 1)- Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates? ► Area of R.

► Radius of inscribed circle in R.

► Distance between two endpoints of R.

► None of these

Question No: 17 (Marks: 1)- Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

► parallel

► perpendicular

► opposite direction

► No relation between them.

Question No: 18 (Marks: 1)- Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ---------► perpendicular

► parallel

► in opposite direction

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Question No: 19 (Marks: 1)- Please choose one

Let

$f\,x\,y(\ ,\ )$ (f_{xx} , f_{yy} *and* f_{xy}) the function has continuous second-order partial derivatives in some

$(x^0, y^0)$and let $D = f^{xx}(x^0, y^0)\, f^{yy}(x^0, y^0) - f^{xy2}(x^0, y^0)$ circle centered at a critical point

If $D > 0$ and $fxx(x0, y0) < 0$ then f has ----------

► **Relative maximum at** $(x0, y0)$

► Relative minimum at $(x0, y0)$

► Saddle point at $(x0, y0)$

► No conclusion can be drawn.

Question No: 20 (Marks: 1)- Please choose one

Which of the following are direction ratios for the line joining the points (1,

3, 5) and (2, −1, 4)? ► 3, 2 and 9

► 1, -4 and -1

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 21 (Marks: 1)- Please choose one

If

$R = R1 \cup R2$ $R1$ $R2$, where and are no overlapping regions then

$\iint_{R1} f(x,y)dA(\ ,\ ) + \iint_{R2} f(x,y)dA(\ ,\ ) =$

► $\int_R f(x,y)dA$

$\iint_{R1} f(x,y)dA(\ ,\ ) \cup \iint_{R2} f(x,y)dA(\ ,\ )$

►

$\iint_R f(x,y)\,dV(\ ,\ )$

▶

$$\iint_{R1} f\,x\,ydA(\ ,\ ) \cap \iint_{R2} f\,x\,ydA(\ ,\ )$$

▶

Question No: 22 (Marks: 1)- Please choose one

If $R = \{(x\ y,\ )/2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, *then*

$\iint_R (4xe^{2y})dA =$

$$\int_0^1 \int_2^4 (4 \quad {}^{2y}) \quad xedydx$$

▶

$$\int_{02}^{14} (4xe^{2y})dxdy$$

▶

$$\int_1^4 \int_0^2 (4 \quad {}^{2y}) \quad xedxdy$$

▶

$$\int_1^4 \int_0^2 (4 \quad {}^{2y}) \quad xedydx$$

▶

Question No: 23 (Marks: 1)- Please choose one http://vustudents.ning.com

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If $R = \{(x\ y,\ )/0 \le x \le 2 \text{ and } 0 \le y \le 3\}$, *then*

$\iint_R (1 - ye^{xy})dA =$

$\int_0^2 \int_0^3 (1 - ye^{xy})dydx$ ▶

$\int_0^2 \int_0^3 (1 - ye^{xy})dydx$ ▶

$\int_2^3 \int_0^0 (1 - ye^{xy})dydx$ ▶

$\int_0^2 \int_2^3 (4 - xe^{2y})dydx$ ▶

Question No: 24 (Marks: 1)- Please choose one

Which of the following is geometrical representation of the equation $y = x^2$, in three dimensional space?

▶ **Parabola**

▶ Straight line

▶ Half cylinder

▶ Cone

Question No: 25 (Marks: 3)

□√3, □ , □ □ □

□ 3 2 □

Consider the point in spherical coordinate system. Find the rectangular coordinatesof this point. http://vustudents.ning.com

Question No: 26 (Marks: 5)

Consider a function $f(x, y) = 4xy - x^4 - y^4$ .One of its critical point is (1, 1) . Find whether (1, 1) is

relative maxima, relative minima or saddle point of $f(x, y)$.

Question No: 27 (Marks: 10)

Find

$f(x, y) = x y^2 - 4y^3$ (2, 1)

Directional derivative of the function, , at the point in the direction of vector,

$$\vec{u} = \frac{\sqrt{3}}{2} i + \frac{1}{2} j$$

MIDTERM EXAMINATION

Spring 2010

MTH301-

Calculus II

Question No: 1 (Marks: 1) - Please choose one

Which of the following is the interval notation of real line?

► $(-\infty , +\infty)$

► $(-\infty , 0)$

► $(0 , +\infty)$

Question No: 2 (Marks: 1) - Please choose one

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

► $y = ax^2 + b \quad (a \neq 0)$

► $x = ay^2 + b \quad (a \neq 0)$

► $y = ax^2 + bx + c \quad (a \neq 0)$

► $x = ay^2 + by + c \quad (a \neq 0)$

http://vustudents.ning.com

Question No: 3 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

► **A point on y-axis**

► Plane parallel to xy-plane

► Plane parallel to yz-axis

► Plane parallel to xz-plane

Question No: 4 (Marks: 1) - Please choose one

Suppose $f(x, y) = x e^{3xy}$. Which one of the statements is correct?

► $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

► $\frac{\partial f}{\partial y} = x^3 exy$

► $\frac{\partial f}{\partial y} = x^4 exy$

► $\frac{\partial f}{\partial y} = x^3 ye^{xy}$

Question No: 5 (Marks: 1) - Please choose one

If

$f(x, y) = x y^2 - y^3 + \ln x$

∂

2

f

t

h

e

n

$\frac{}{\partial x^2}$

=

$2xy + \frac{1}{x^2}$ ▶

$2y + \frac{1}{x^2}$ ▶

$2xy - \frac{1}{x^2}$ ▶

$2y - \frac{1}{x^2}$ ▶

Question No: 6 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$$\frac{\partial w}{\partial r} =$$

$$\frac{\partial w}{\partial x}\frac{x\partial}{\partial r} + \frac{\partial w}{\partial y}\frac{y\partial}{\partial r} + \frac{\partial w}{\partial z}\frac{z\partial}{\partial r}$$

$$\frac{\partial}{\partial r}\frac{\partial w}{\partial x}\frac{x}{r}\frac{\partial}{\partial r} + \frac{\partial w}{\partial y}\frac{y}{r}\frac{\partial}{\partial r} + \frac{\partial w}{\partial z}\frac{z}{r}\frac{\partial}{\partial r}$$

$$\frac{\partial w}{\partial x}\frac{x}{\partial r}\frac{\partial}{s} + \frac{\partial w}{\partial y}\frac{y}{\partial r}\frac{\partial}{\partial s} + \frac{\partial w}{\partial z}\frac{z}{\partial r}\frac{\partial}{s}$$

$$\frac{\partial w}{\partial r}\frac{r\partial}{\partial x} + \frac{\partial w}{\partial r}\frac{r\partial}{\partial y} + \frac{\partial w}{\partial r}\frac{r\partial}{\partial z}$$

Question No: 7 (Marks: 1) - Please choose one

Is

$f(x, y)$ the function continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \dfrac{3x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$$

$f(x, y)$ is continuous at origin

$f(0, 0)$ **is not defined**

$$\lim f(x, y)$$

$f(0, 0)$ $(x, y,) \to (0, 0)$

► **is defined but** **does not exist**

$\lim_{(x, y,) \to (0, 0)} f(x, y,)$ $f(0, 0)$

► is defined and exists but these two numbers are not equal.

Question No: 8 (Marks: 1) - Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

► Area of R.

► Radius of inscribed circle in R.

► Distance between two endpoints of R.

► None of these (not sure)

http://vustudents.ning.com

Question No: 9 (Marks: 1) - Please choose one

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

► Parallel

► Perpendicular

► In opposite direction

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

► perpendicular

► parallel

► in opposite direction

Question No: 11 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives

(*fxx , fyy and fxy*) in some circle centered at a critical point (*x*0, *y*0)and let
D = fxx(x0, y0) fyy(x0, y0) − fxy2(x0, y0)

If *D* > 0 and *fxx*(*x*0, *y*0) > 0 then *f* has ----------

► Relative maximum at (*x*0, *y*0)

► Relative minimum at (*x*0, *y*0)

► Saddle point at (*x*0, *y*0)

► No conclusion can be drawn. http://vustudents.ning.com

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives

(*fxx , fyy and fxy*) in some circle centered at a critical point (*x*0, *y*0)and let
D = fxx(x0, y0) fyy(x0, y0) − fxy2(x0, y0)
If *D* > 0 and *fxx*(*x*0, *y*0) < 0 then *f* has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives

(*fxx , fyy and fxy*) in some circle centered at a critical point (*x*0, *y*0)and let
D = fxx(x0, y0) fyy(x0, y0) − fxy2(x0, y0)

If $D < 0$ then f has ----------

► Relative maximum at (*x*0, *y*0)

► Relative minimum at (*x*0, *y*0)

► **Saddle point at** (*x*0, *y*0)

► No conclusion can be drawn

Question No: 14 (Marks: 1) - Please choose one

Let $(x1, y1, z1)$ and $(x2, y2, z2)$ be any two points in three dimensional space. What does the formula $\sqrt{(x2 - x1)2 + -(y2y1)2 + -(z2z1)2}$ calculates?

► **Distance between these two points**

► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 15 (Marks: 1) - Please choose one

The

---

function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- ► $x \neq y$
- ► $x \leq y$
- ► $x > y$ http://vustudents.ning.com

Question No: 16 (Marks: 1) - Please choose one

Plane is an example of ----------

- ► Curve
- ► Surface
- ► Sphere
- ► Cone

Question No: 17 (Marks: 1) - Please choose one

If $R = R1 \cup R2$, where $R1$ and $R2$ are no overlapping regions then $\iint_{R1} f(x, y)dA + \iint_{R2} f(x, y)dA =$

$\iint_R f(x,y)\,dA(\ ,\ )$

▶

$\iint_{R1} f(x,y)\,dA(\ ,\ ) \cup \iint_{R2} f(x,y)\,dA(\ ,\ )$

▶

$\iint_R f(x,y)\,dV(\ ,\ )$

▶

$\iint_{R_1} f(x,y)\,dA(\ ,\ ) \cap \iint_{R_2} f(x,y)\,dA(\ ,\ )$

http://vustudents.ning.com

▶

Question No: 18 (Marks: 1) - Please choose one

$If\ R = \{(x,y,)/0 \le x \le 2 and 1 \le y \le 4\}, then$

$\iint_R (6x^2 + 4xy\,dA^3) =$

$\int_1^4 \int_0^2 (6x^2 + 4xy\,dydx^3)$

▶

$\int_2^4 \int_0^1 (6x^2 + 4xy\,dxdy^3)$

▶

$$\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dxdy$$

▶

$$\int_{2}^{4}\int_{0}^{1}(6x^2+4xy^3)dxdy$$

▶

Question No: 19 (Marks: 1) - Please choose one

If $R=\{(x\,y,\,)/0\le x\le 2 \text{ and } 0\le y\le 3\}$, *then*

$\iint_R (1-ye^{xy})dA =$

$$\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$$

▶

$$\int_{0}^{2}\int_{0}^{3}-(1ye^{xy})dxdy$$

▶

$$\int_{2}^{3}\int_{0}^{0}-(1ye^{xy})dxdy$$

▶

$$\int_{0}^{2}\int_{2}^{3}(4^{2y})xedydx$$

▶

Question No: 20 (Marks: 1) - Please choose one

$If R = \{(x,y) / \quad 0 \le x \le 4 \, and \, 0 \le y \le 9\}, then$

$\int_R (3x - 4x\sqrt{y} dA) =$

http://vustudents.ning.com

$\int_0^9 \int_0^4 (3x - 4x\sqrt{y}) dx$

►

$\int_0^4 \int_0^9 (3x - 4x\sqrt{y}) dx$

►

$\int_0^9 \int_0^4 (3x - 4x\sqrt{y}) dx$

►

$\int_0^4 \int_0^9 (3x - 4x\sqrt{y}) dy$

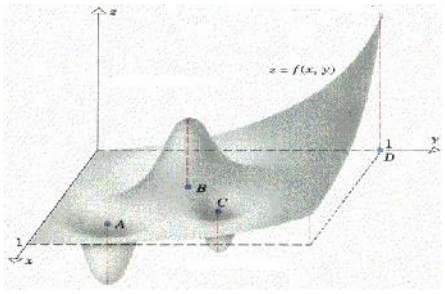
►

$\int_0^4 \int_0^9 (3x - 4x\sqrt{y}) dy dx$

►

Question No: 21 (Marks: 2)

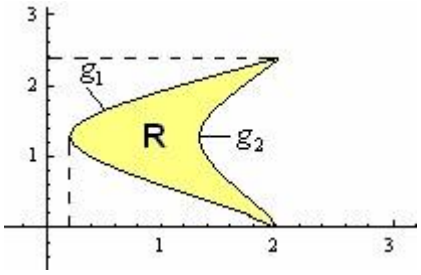
Following is the graph of a function of two variables



In its whole domain, state whether the function has relative maximum value or absolute maximum value at point B. Also, justify your answer

Question No: 22 (Marks: 2)

Let the function $f(x, y)$ is continuous in the region R, where R is bounded by graph of functions g_1 and g_2 (as shown below). http://vustudents.ning.com



In the following equation, replace question mark (?) with the correct value.

$$\iint_R f(x, y)\, dA = \int_{?}^{?} \int_{?}^{?} f(x, y)\, ____?____$$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$$\iint (3 + -2x \quad 3y^2)\, dx\, dy$$

Question No: 24 (Marks: 3)

Let $f(x, y, z)$

$= yz^3 - 2x^2$ *Find*

the gradient of

f .

Question No: 25 (Marks: 5)

Find, Equation of Normal line (in parametric form) to the surface

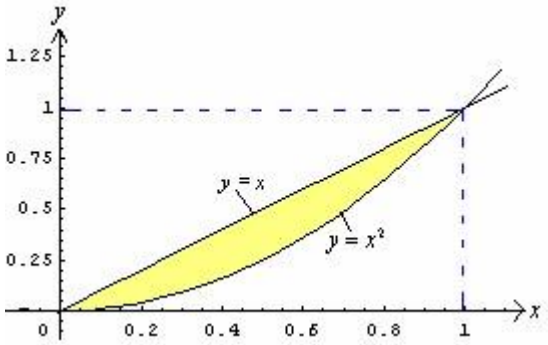
$f x y z(, ,) = xy + 2yz - xz^2 + 10$ (5− , 5, 1)

at the point

Question No: 26 (Marks: 5)

Use double integral in rectangular co-ordinates to compute area of the region bounded

$y = x$ *and* $y = x^2$ by the curves , in the first quadrant.



MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 3)

MTH301

Suppose $f(x, y) = x e^{3xy}$. Which one of the following is correct?

$\frac{\partial f}{\partial x} = 3x e^{2xy} + x y e^{xy}$

$\frac{\partial f}{\partial x} = 3$

x

y

e

x

y

∂

x

$\frac{\partial f}{\partial x} = 3x e^{2xy} + x e^{4xy}$

$\frac{\partial f}{\partial x} = 3x e^{2xy}$

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Area of R.

Radius of inscribed circle in R.

Distance between two endpoints of R. None of these

What is the distance between points (3, 2, 4) and (6, 10, -1)?

$7\sqrt{2}$

$2\sqrt{6}$

$\sqrt{34}$

$7\sqrt{3}$

---------- planes intersect at right angle to form three dimensional space.

Three

4

8

12

There is one-to-one correspondence between the set of points on co-ordinate line and ----------**Set of real numbers**

Set of integers

Set of natural numbers

Set of rational numbers

f x y(,)

Let the function has continuous second-order partial derivatives

(*fxx , fyy and fxy*) in some circle centered at a critical point ($x\ y0$, 0)and let
$D = fxx(x0, y0)\ fyy(x0, y0) - fxy2(x0, y0)$

If $D = 0$ then ----------

f has relative maximum at ($x\ y0$, 0)

f has relative minimum at ($x\ y0$, 0)

f has saddle point at ($x\ y0$, 0)

No conclusion can be drawn.

If $R = \{(x\ y,)/0 \leq \leq x \quad 2 \text{ and } 0 \leq \leq y \ 3\}$, *then*

$\iint (1 - ye^{xy})dA =$
R

$$\int_0^2 \int_0^3 (1 - ye^{xy})\, dydx$$

$$\int_0^2 \int_0^3 (1 - ye^{xy})\, dydx$$

$$\int_2^3 \int_0^0 (1 - ye^{xy})\, dydx$$

$$\int_0^2 \int_2^3 (4 - xe^{2y})\, dydx$$

Suppose $f(x, y) = 2xy$ *where* $x = +t^2 - 1$ *and* $y = -3t$. Which one of the following is true?

$\frac{df}{dt} = -(-6t)^2 - 4t - 2$

$\frac{df}{dt} = -6t - 2$

$\frac{df}{dt} = 3$

$= + -4t$

$6t$

6

$\frac{df}{dt} = -(-6t)^2 + 12t - 2$

Let i, j and k be unit vectors in the direction of x-axis, y-axis and z-axis respectively. Suppose

$\rightarrow$

$a = +-2i\ 5\ j \quad k \quad \vec{}_a$ that . What is the magnitude of vector ?

6

30

$\sqrt{30}$

$\sqrt{28}$

A straight line is ---------- geometric figure.

One-dimensional

Two-dimensional

Three-dimensional

Dimensionless

$If\ R = \{(x\ y,\ )/0 \leq \leq x \quad 2 and 1 \leq \leq y \quad 4\}, then$

$\iint (6x^2 + 4xy\ dA^3) =$

R

4 2

$\int\int (6x^2 + 4xy\ dydx^3)$

1 0

2 4

$\int\int (6x^2 + 4xy\ dxdy^3)$

0 1

4 2

$\int\int (6x^2 + 4xy\ dxdy^3)$

1 0

---

4 1

$\int\int (6x^2 + 4xy\ dxdy^3)$

2 0

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent□ □ □ □

a b and c,

edges formed by the vectors

? □ □ □

$|a\ b\ c \times \times ( )|$

□ □□□

$|a\ b\ c \square\square(|\square\square)$

□ □ □

$a\ b\ c \square\square(\times)$

□ □□□

$|a\ b\ c^{\times}(\square\square)|$

$f\ x\ y(,\ ) = \sqrt{-y\ \ x}$

The function is continuous in the region --------- and discontinuous elsewhere. $x\ y \neq$

$x \leq y$

$x\ y>$

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

paralle l

perpen dicular

opposit e directi on

No relation between them.

$f\ x\ y(,) = x\ e^{3\ xy}$

Suppose . Which one of the statements is correct?

$\frac{\partial f}{=} 3$

3 x e ^{x} ^{y}

$\frac{\partial}{y}$

∂f 3

$$\frac{\partial}{\partial y} = x e^{xy}$$

$$\frac{\partial f}{\partial y} = x e^{4xy}$$

$$\frac{\partial f}{\partial y} = x^3 y e^{xy}$$

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

perpendicular

parallel in opposite direction

f x y(,)

Let the function has continuous second-order partial derivatives

(f_{xx} , f_{yy} *and* f_{xy}) in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Relative maximum at (x_0, y_0)

Relative minimum at (x_0, y_0)

Saddle point at

No conclusion can be drawn.

If $R = \{(x\ y,\ )/0 \leq \leq x \quad 2 \text{ and } -1 \leq \leq y\ \ 1\}$, *then*
$\iint (x + 2y\ dA^2) =$
R

1 $2\int \int (x + 2y$
$dydx^2)$
$-1\ 0$

2 $-1 \int \int (x +$
$2y\ dxdy^2)$
0 1

1 2
$\int \int (x + 2y\ dxdy^2)$
$-1\ 0$

2 0 $\int \int (x + 2y$
$dxdy^2)$
1 -1

$$f\ x\ y\ z(\ ,\ ,\ ) = \frac{x\ y^2}{\quad} + xyz$$

z If then what is the value of f (1, 1, 1) =1

f (1, 1, 1) = 2

$$f(1, 1, 1)? f(1, 1, 1) = 3 f(1, 1, 1) = 4$$

If $R = \{(x\ y,\ )/0 \le \le x \quad 4$ *and* $0 \le \le y\ 9\}$, *then*

$\iint_R (3x - 4x\ xy\sqrt{dA}) =$

$$\int_0^9 \int_0^4 (3x - 4x\ xy\ d\sqrt{ydx})$$

$$\int_0^4 \int_0^9 (3x - 4x\ xy\ d\sqrt{xdy})$$

$$\int_4^9 \int_0^0 (3x - 4x\ xy\ d\sqrt{xdy})$$

$$\int_0^9 \int_0^4 (3x - 4x\ xy\ dydx)$$

Let $f\ x\ y(\ ,\ ) = 2 + x + \frac{y^2}{4}$

Q- *Find the gradient of f*

2MARKS

Q - Let the function $f\ x\ y(\ ,\ )$ is continuous in the region R, where R is a rectangle as shown below.complete the following equation

$\iint_R$ f x y dA(,) = $\int\int$ f x y(,) _____

2MARKS

Q.Find all critical points of the function $f\,x\,y(\,,\,) = 4xy - x^3 - 2y^2$

$$\int_{}^{4}\int_{}^{2}(6\ ^{2}+4\ \ ^{3})_{}xxy\,dx\,dy$$

Evaluate

Q-Evaluate the following double integral.

$\iint(3+ -2x \quad 3y^2)\,dx\,dy$

3MARKS

$$y = \frac{1}{x}$$

$$2$$

$$x$$

Q- Let . If changes from 3 to 3.3, find the approximate change in the value of y using differential dy.

3MARKS

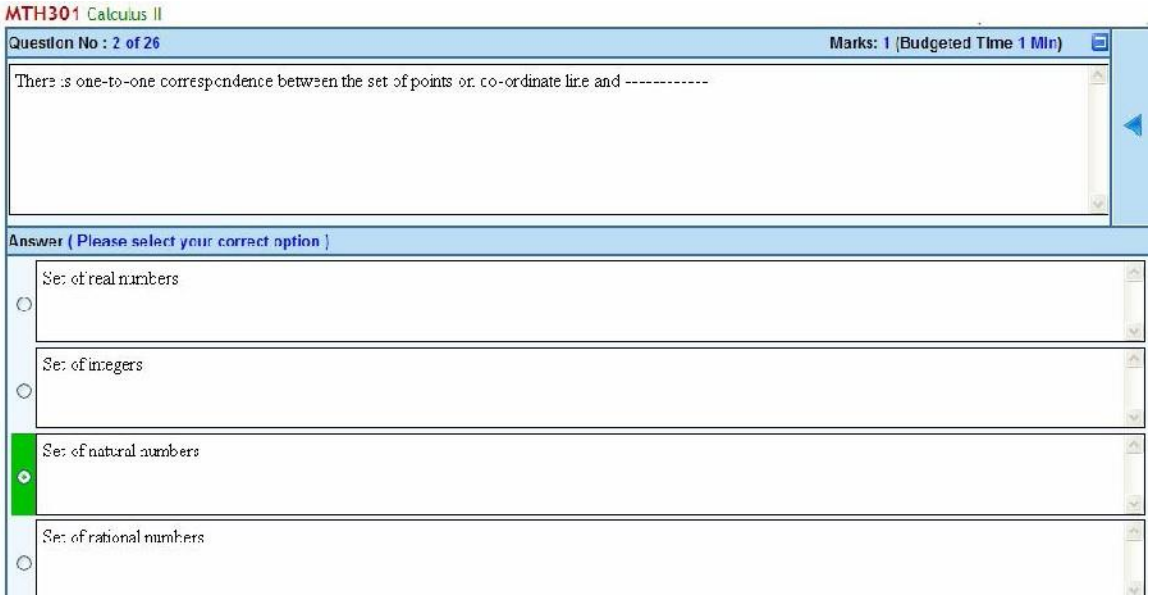
MIDTERM EXAMINATION

MTH301

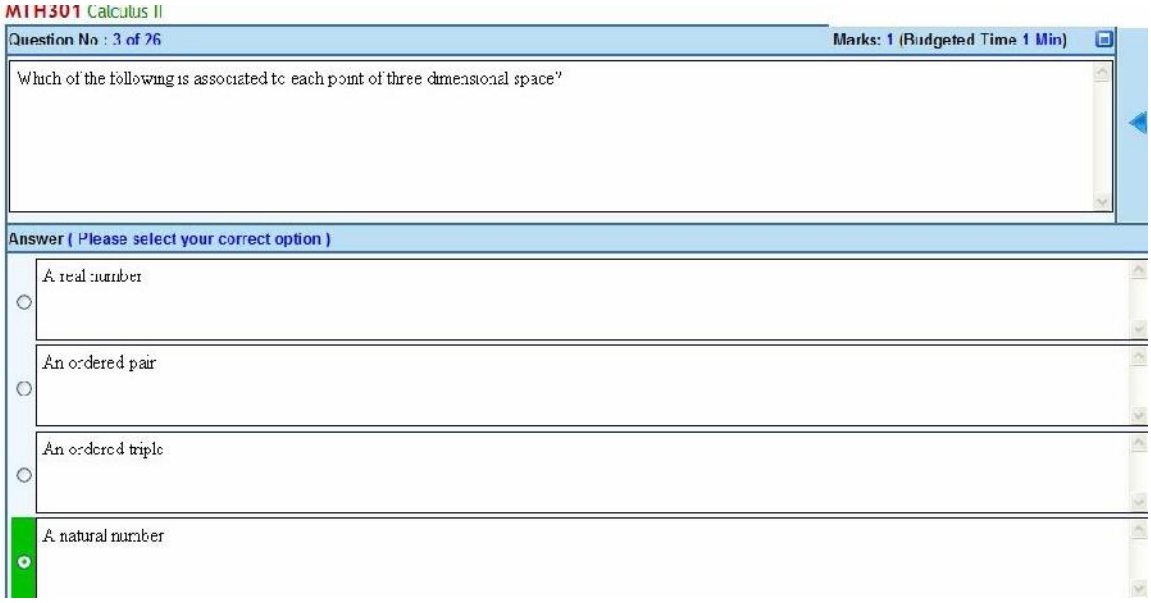
That all is collected by removing the mistakes of the one paper file, hope it helpful…!



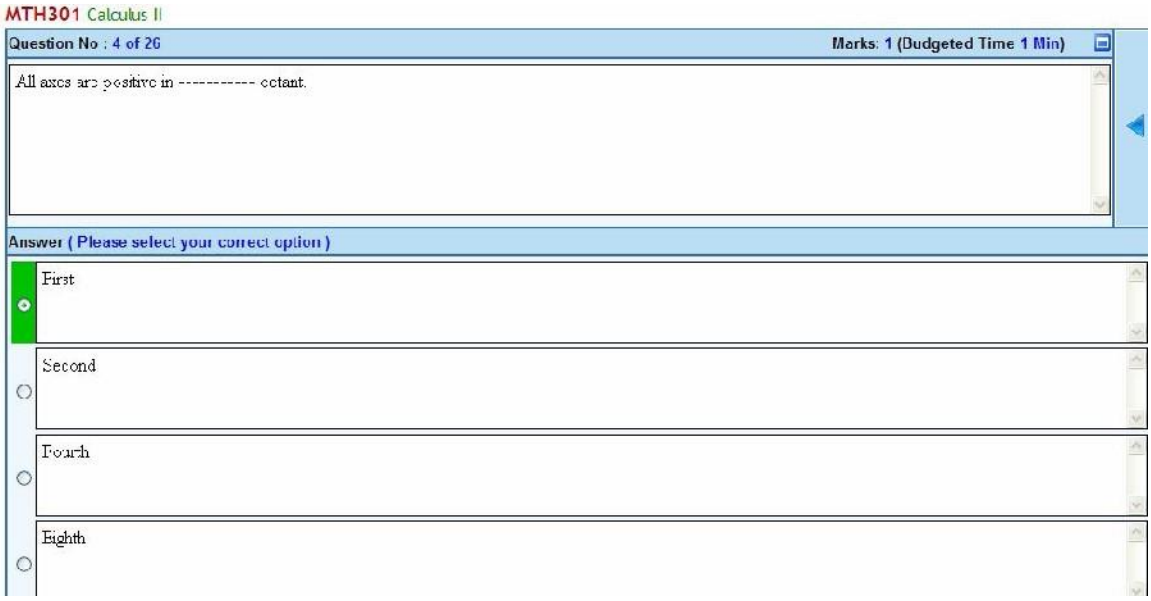
In that the answer is “unique point”



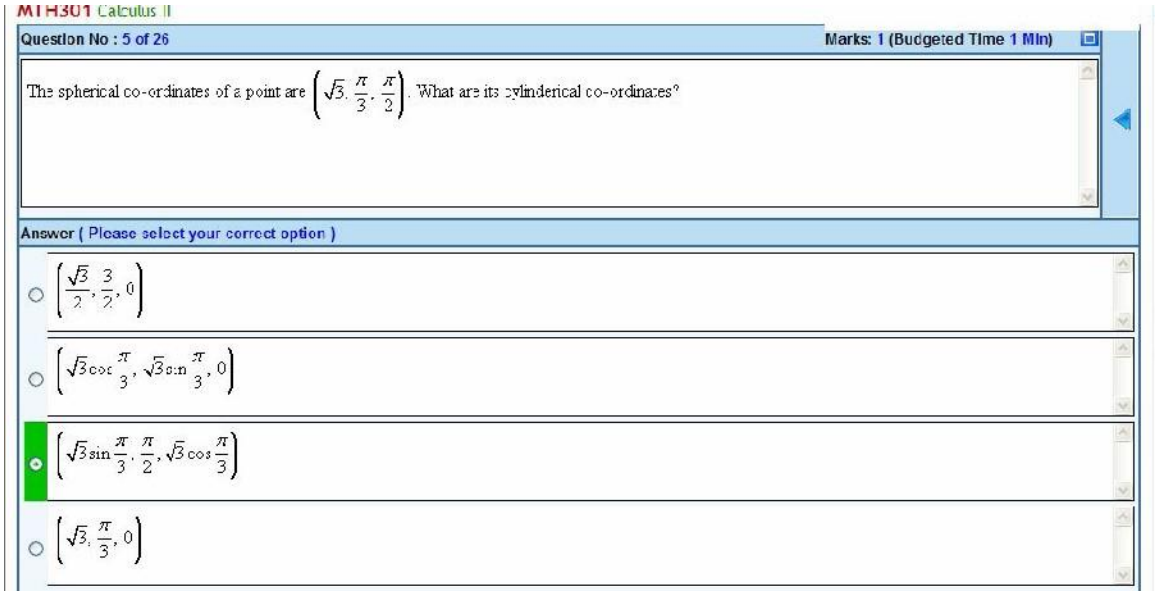
In that the correct answer is the “Set of Real Numbers”.



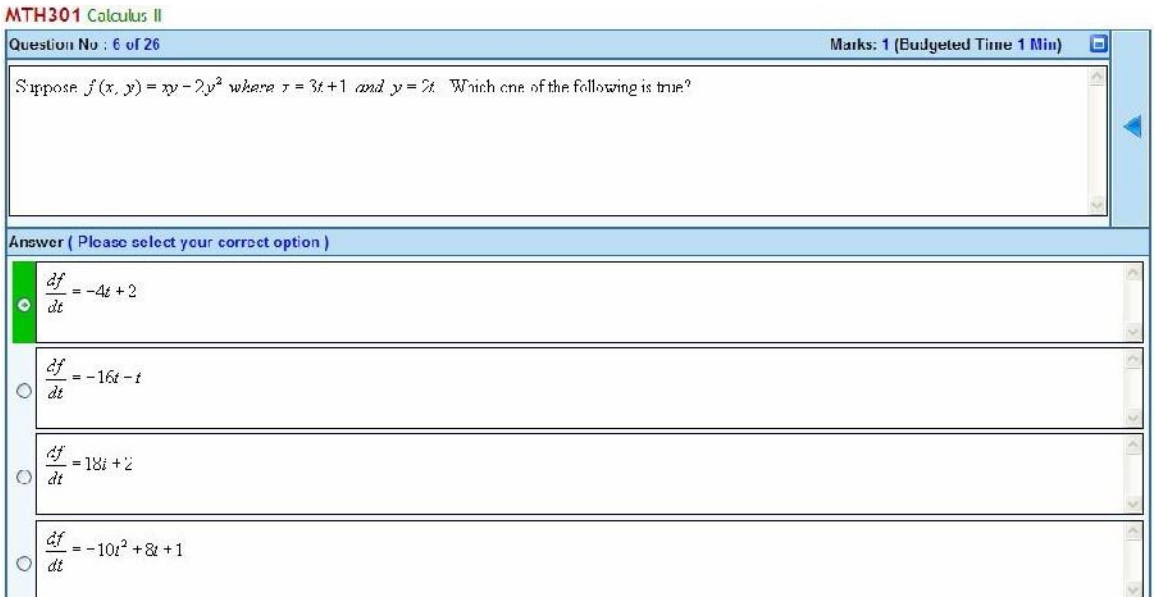
In this the exact answer is “an ordered triple”.



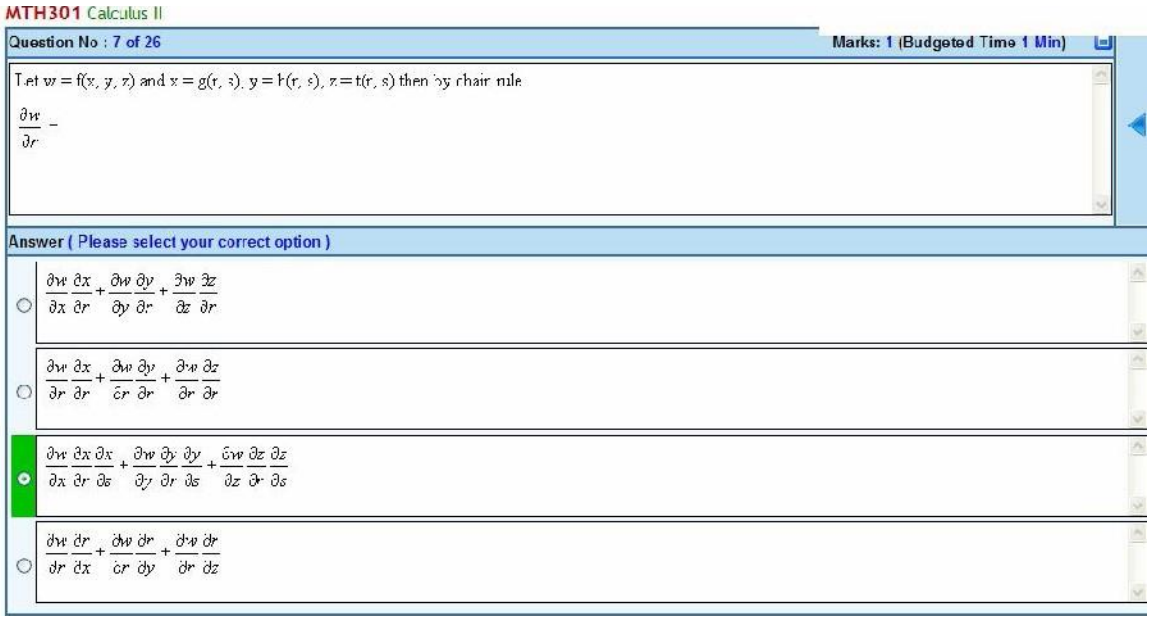
It's right



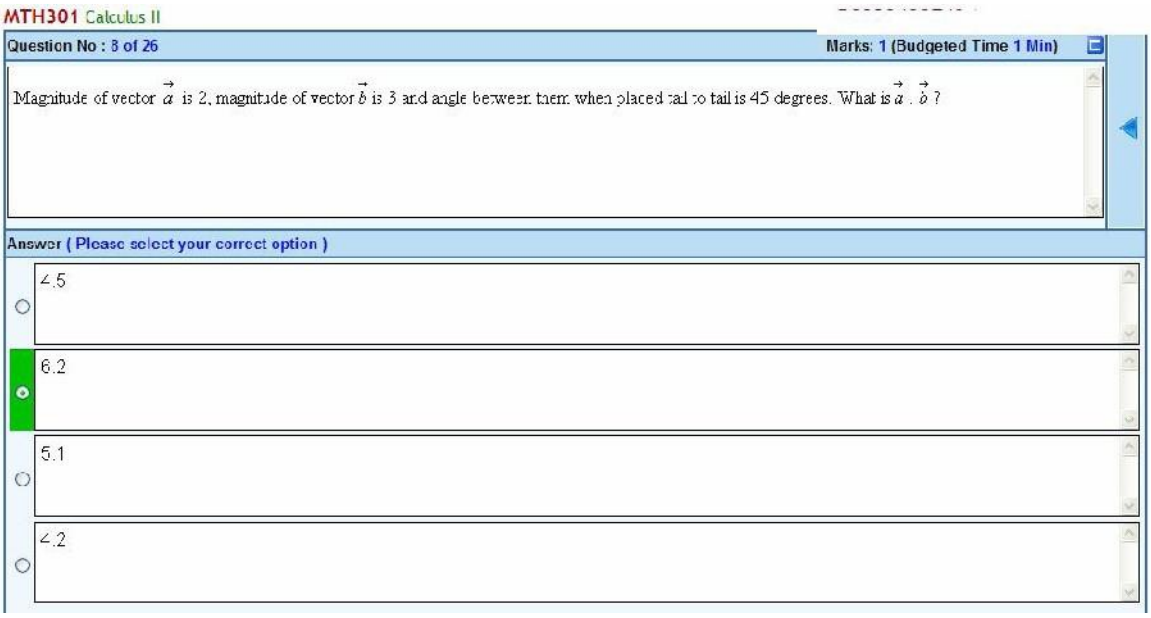
This one is also wrong as the answer is "4th one".because the central point of spherical is pie/3 that should be same in cylindrical so, that's only occur in last one…



That's right



Here's answer is the "2nd" one…!

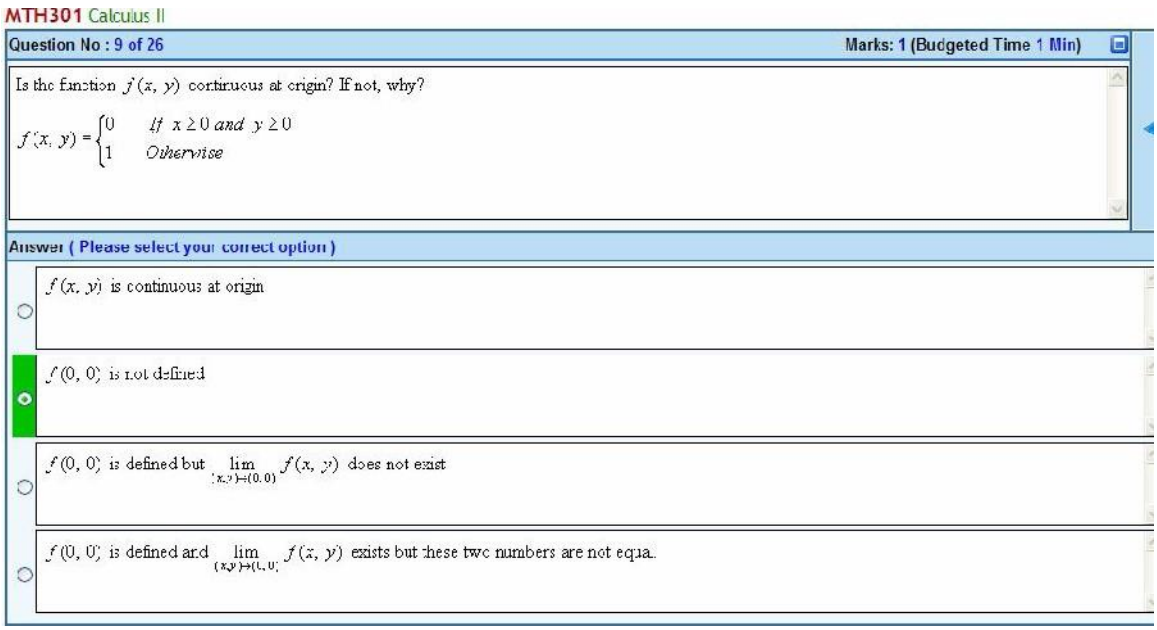


That can't say exactly guys, its formula is following…!

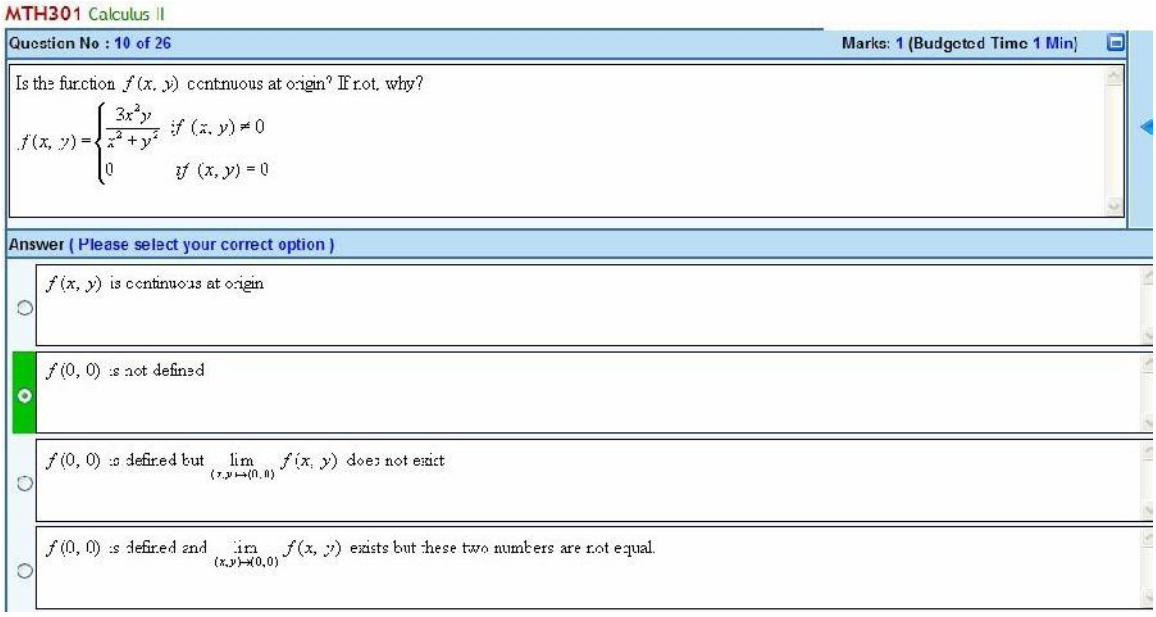
a.b = |a||b| cosØ

Ø=angle between the

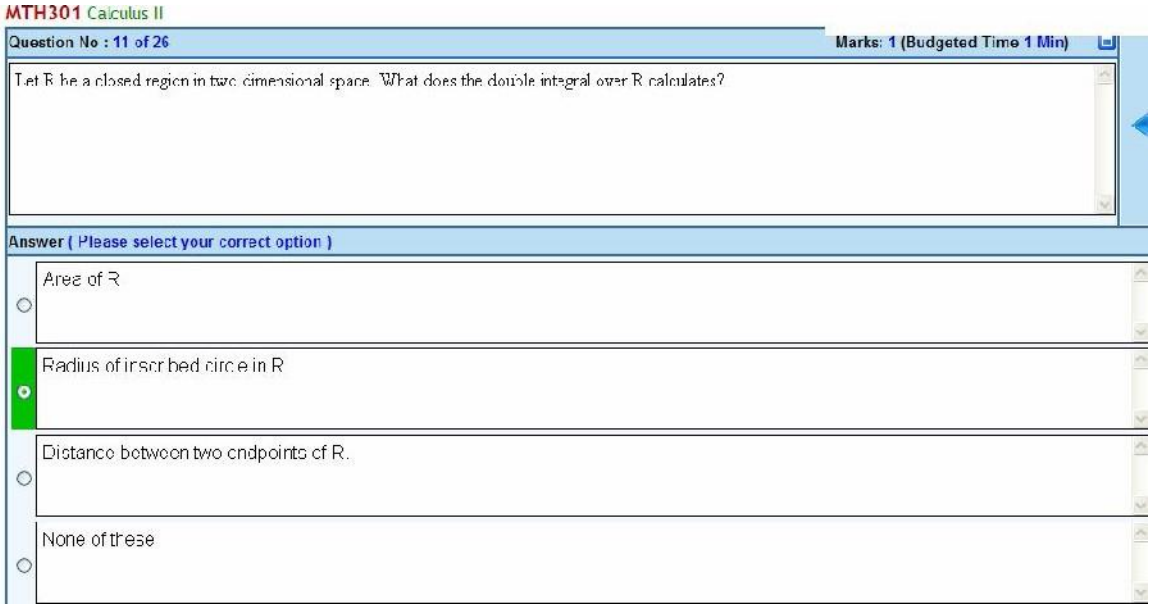
vectors



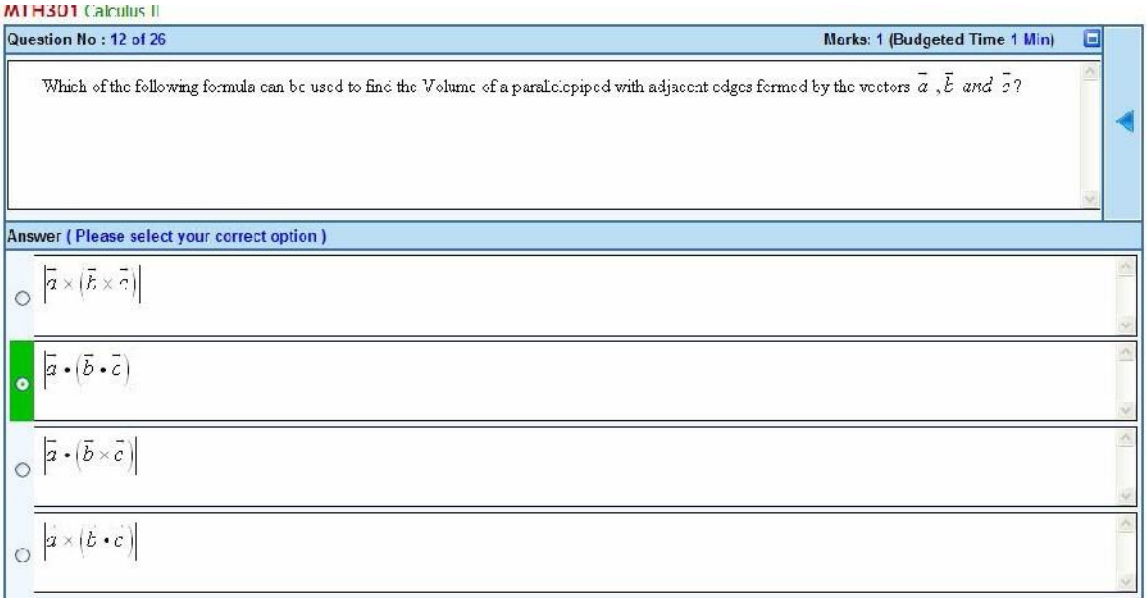
That's correct answer…!



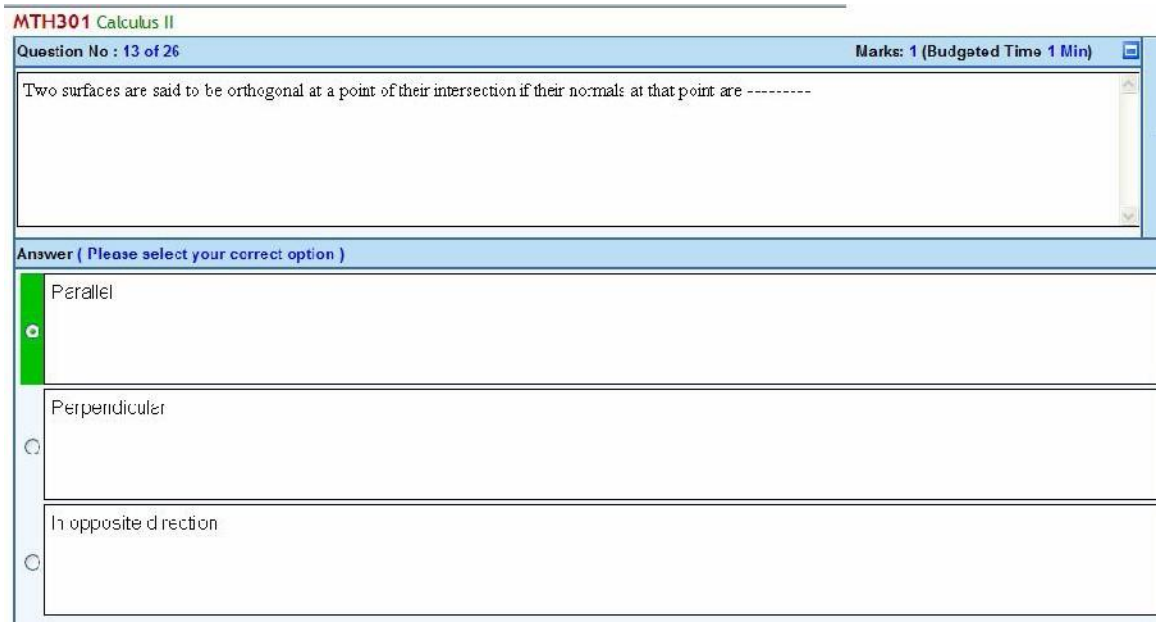
But that's wrong the correc one is the 1st..



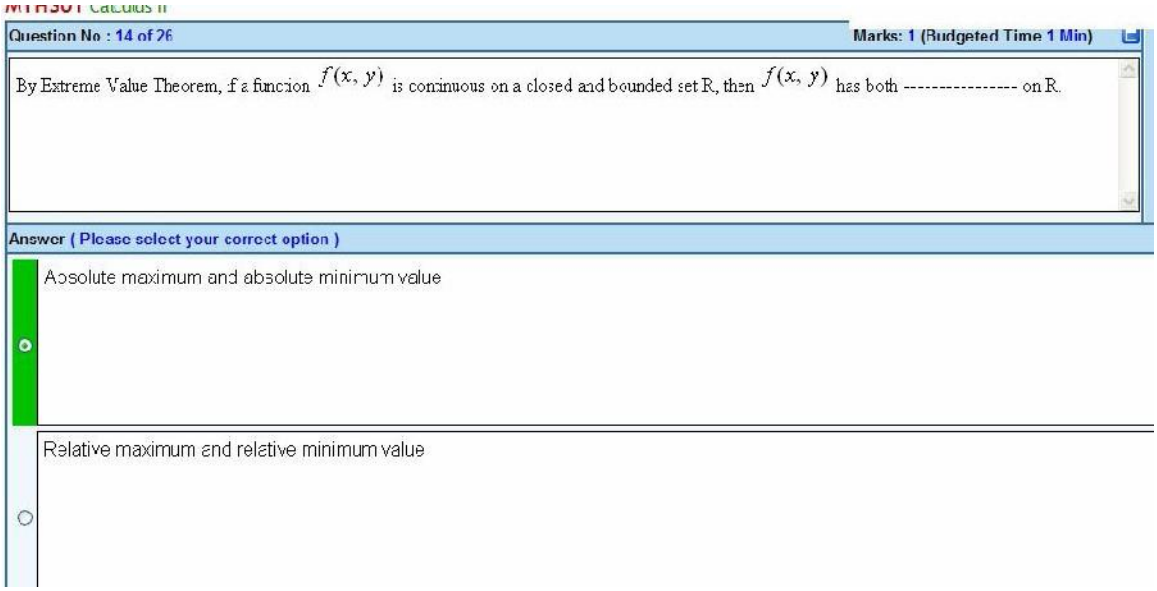
That's the first answer, as that was been define in the lecture 22…!



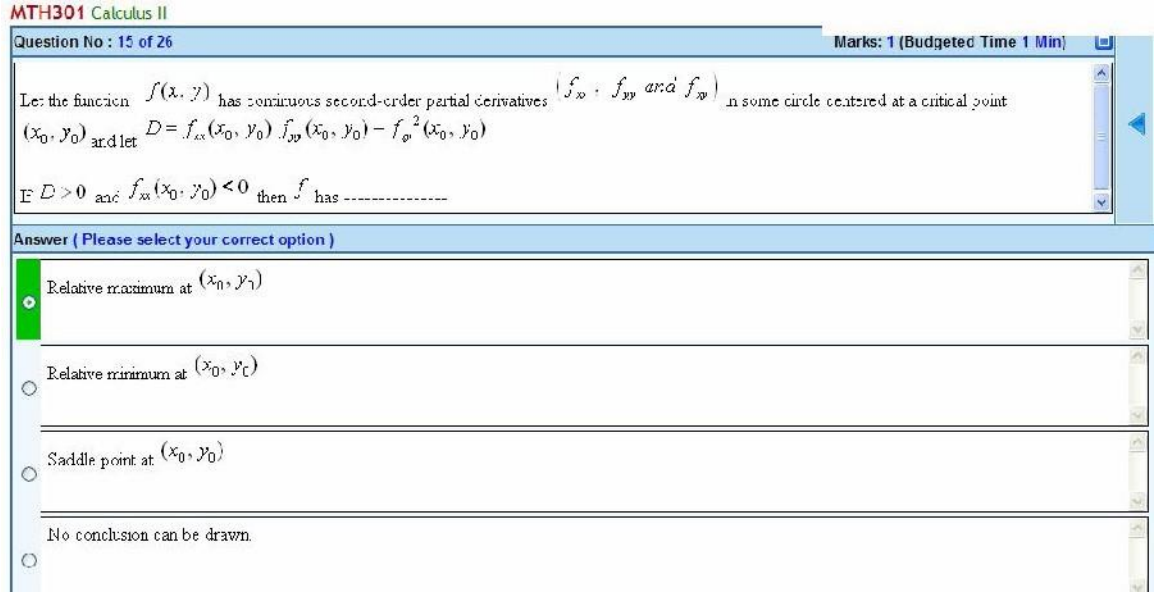
Here's the answer is the last one, you can check that on the start of lecture 11th, handout…!



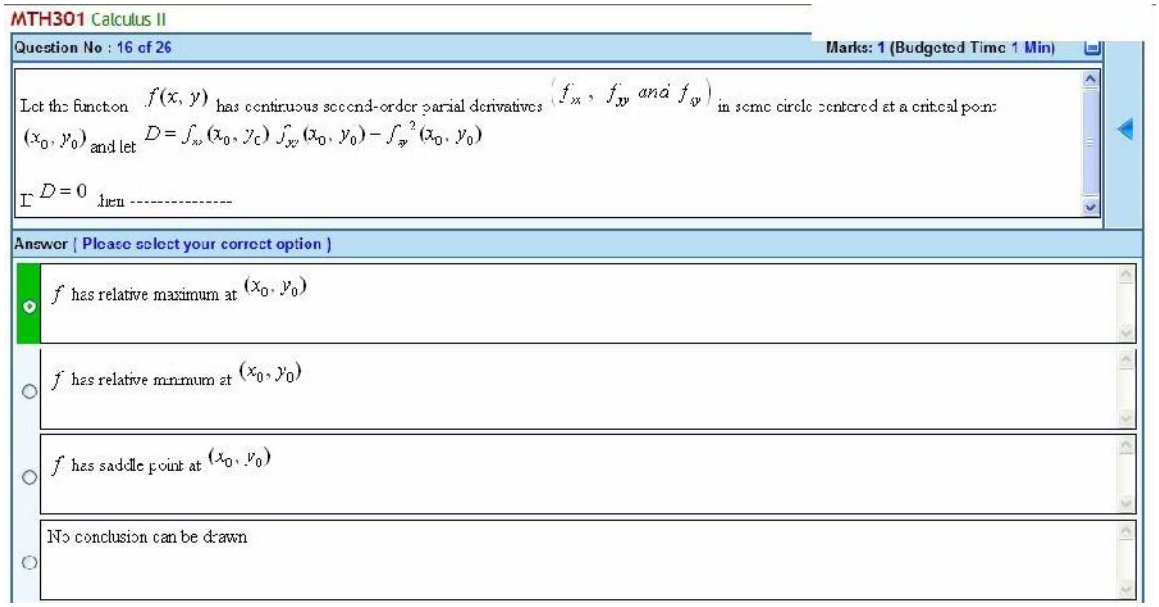
Its answer is the "perpendicular"…!



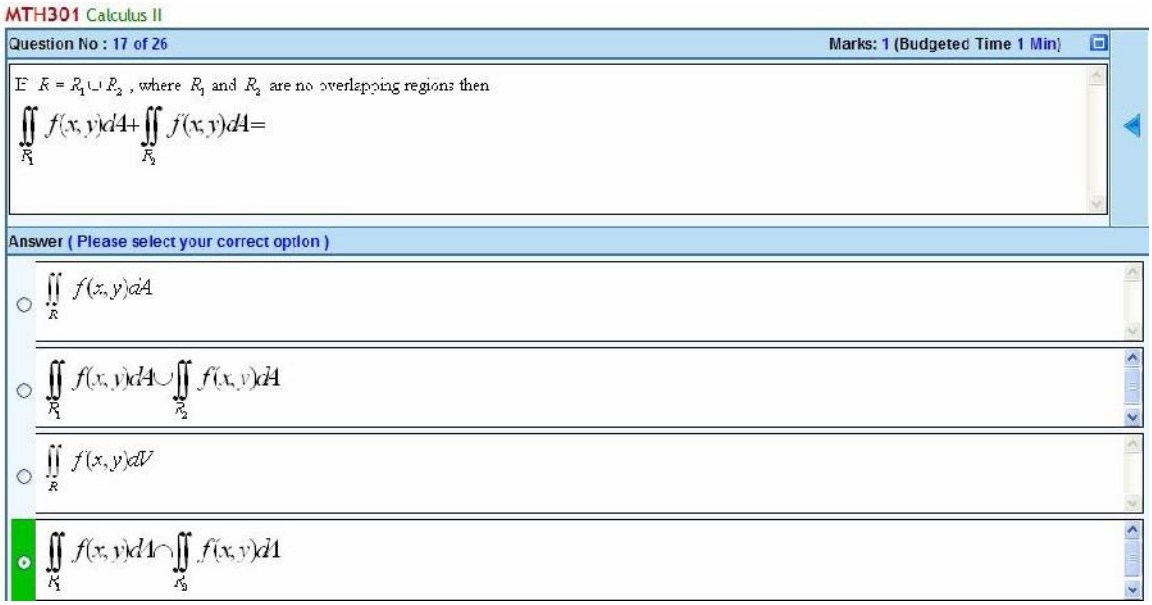
That's right answer…!



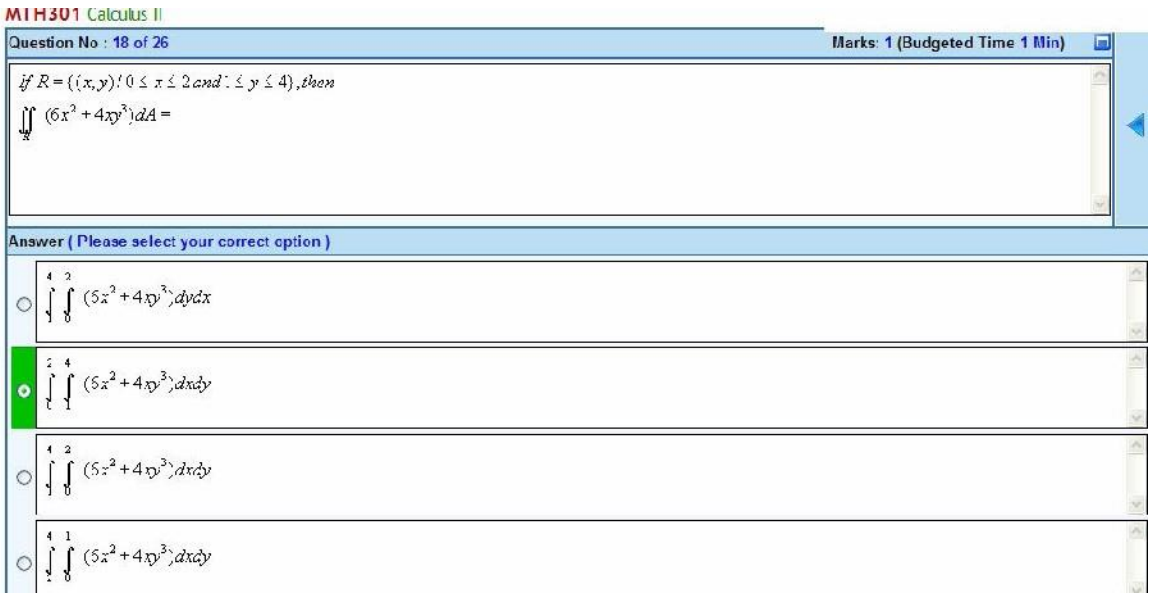
This is also right one..!



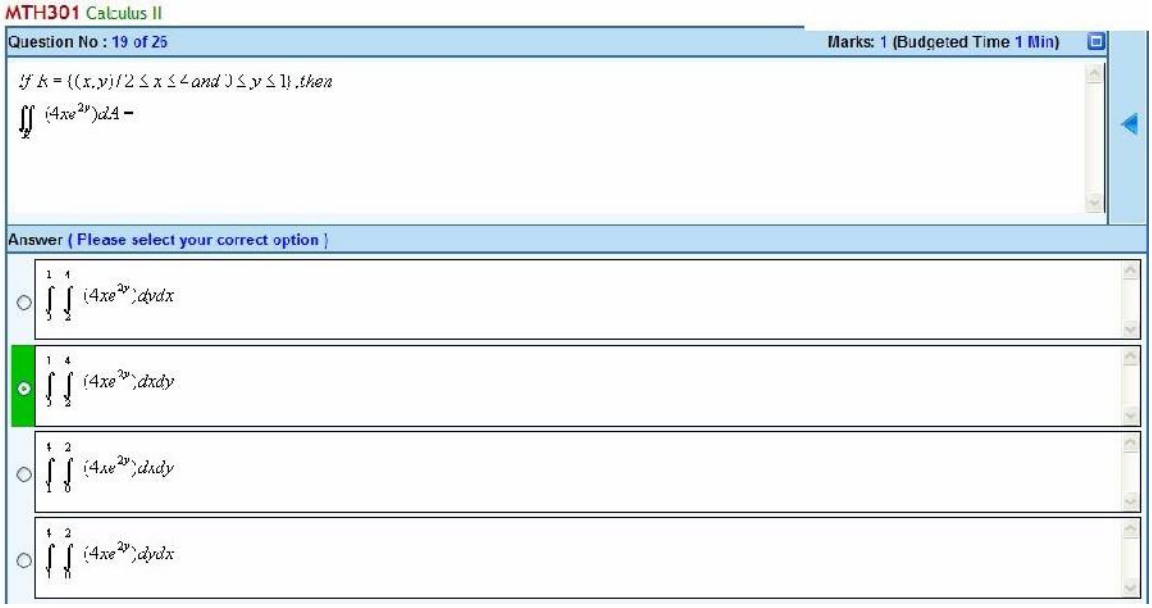
"No conclusion can be drawn" is the answere…!



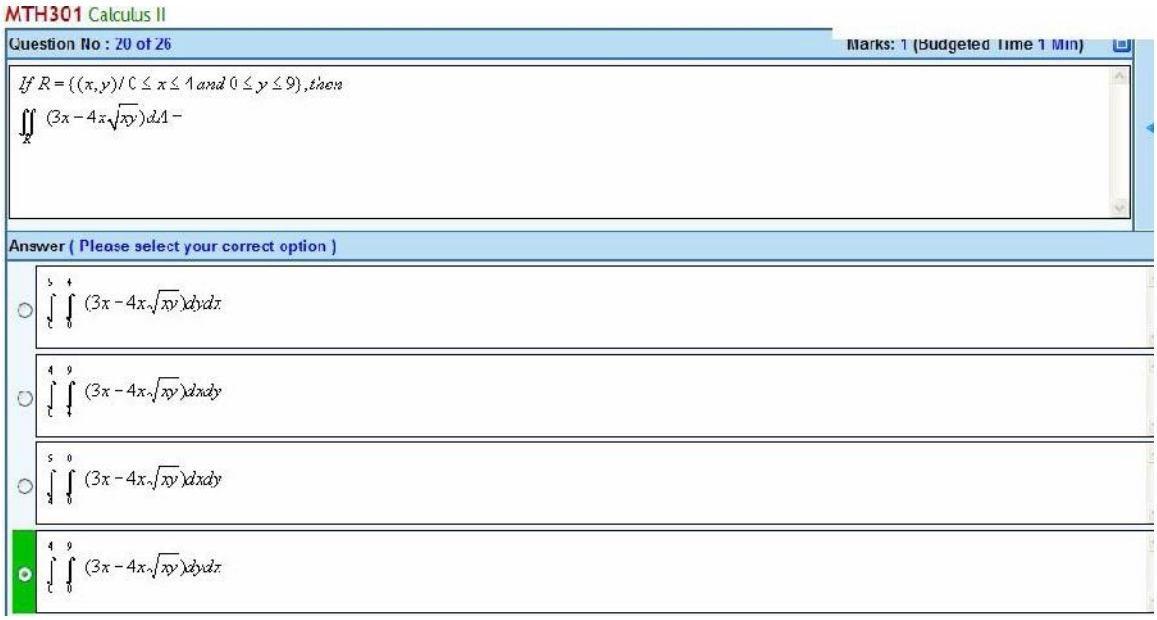
Its answer is the first one option…!



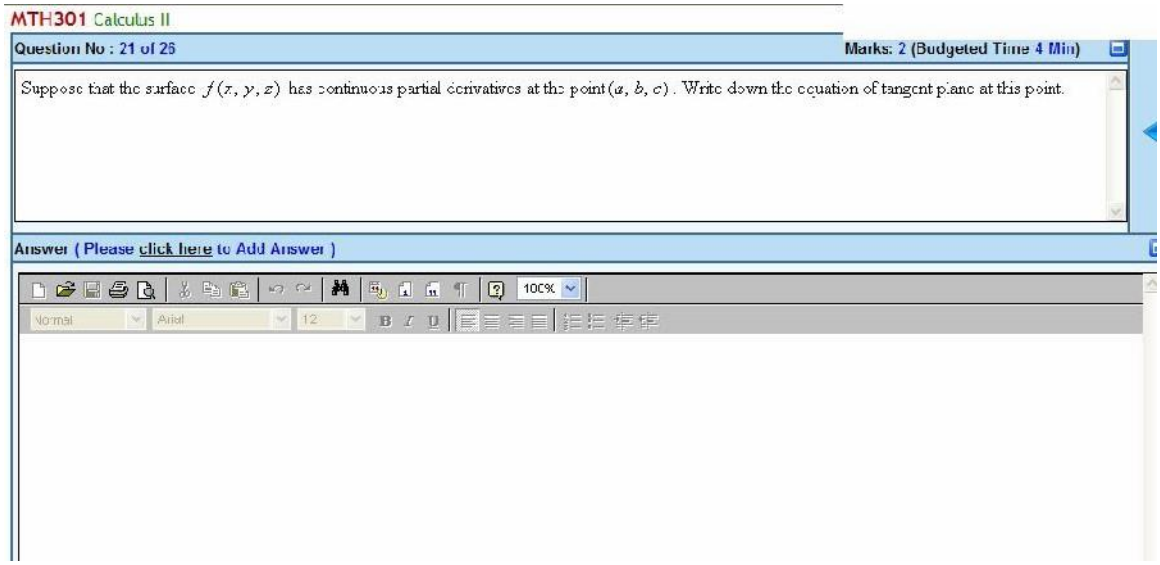
Its answer is the 3rd option



It's right one…!

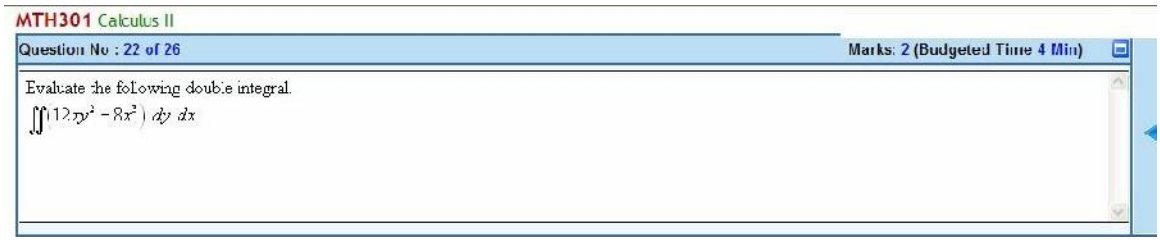


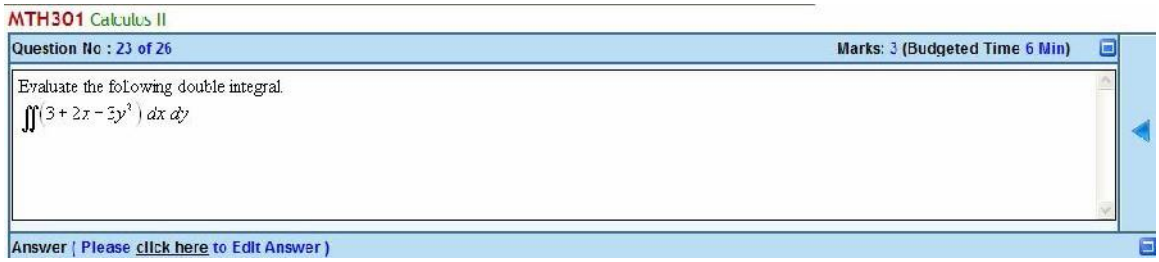
This one's also right…



Its equation is

$$f_a\, x_x(\,)(-x_0)+ f_b\, y_y(\,)(-y_0)+ f_c\, z_z(\,)(-z_0) = 0$$





These both can be solve by the following example the me gonna paste…!

Evaluate the following double integral.

$\iint(3+ -2x \quad 3y^2)\, dx\, dy$

Solution:

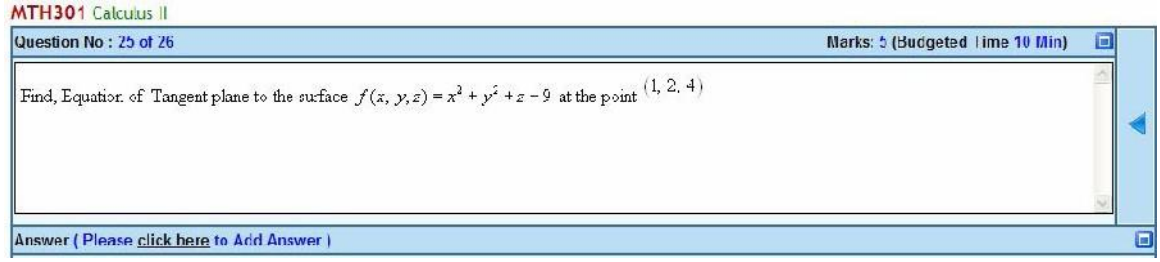
$\iint(3+ 2x -3y^2)\, dx\, dy$

$\int(3x + 2x^2 -3xy\, dy^2)$

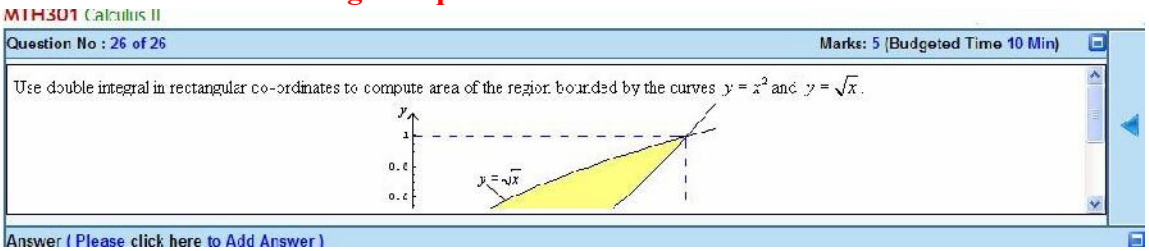
$3xy + 2x\, y^2 - xy^3$



I **didn't got this question dear's…!**



You can use the above tangent equation to sove this…!



Its answer is the under mentioned…!

as our require is area of the following region

$V = \iint_R 1\,dA = \iint_R dA$ we can compute area by keeping f(x,y) const

$$= \int_0^1 \int_{x^2}^{x} dy\,dx$$

$$= \int_0^1 [x]_{x^2}^{x}$$

$$= \int_0^1 (x - x^2)$$

$$= [x\,\frac{x^2}{2} - \frac{x^3}{3}]_0^1$$

$$= \frac{1}{2} - \frac{1}{3}$$

$$= \frac{3-2}{2}$$

$$= \frac{1}{6}$$

$$\frac{}{6}$$

$$= \frac{1}{6}$$ that's exactly what we require...!

MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 3)

Time: 60 min

Question No: 1 (Marks: 1) - Please choose one

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► **A real number**

► A rational number

► A natural number

Question No: 2 (Marks: 1) - Please choose one

If $a > 0$, then the parabola $y = ax^2 + bx + c$ opens in which of the following direction?

► **Positive x - direction**

► Negative x - direction

► Positive y - direction

► Negative y - direction

Question No: 3 (Marks: 1) - Please choose one

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

$\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{2}\right)$

►

$\left(2\sqrt{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right)$

►

$\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

►

$\left(\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

►

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit ---------- exist at that point.

□□□□□□□▶□Always

□□□□□□□▶□Never

□□□□□□□▶□**May**

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x,\ y) = x^3 e^{xy}$. Which one of the statements is correct?

□□□□□□□▶□ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^3 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 6 (Marks: 1) - Please choose one

If $f(x,\ y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

$2xy + \frac{1}{x^2}$

□□□□□□□▶□

$2y + \frac{1}{x^2}$

□□□□□□□▶□

$2xy - \frac{1}{x^2}$

□□□□□□□▶□

$2y - \frac{1}{x^2}$

□□□□□□□▶□

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y^2 \ \ where \ x = 3t + 1 \ \ and \ \ y = 2t$. Which one of the following is true?

$\frac{df}{dt} = -4t + 2$

□□□□□□□▶□

$\frac{df}{dt} = -16t - t$

□□□□□□□▶□

$\frac{df}{dt} = 18t + 2$

□□□□□□□▶□

$\frac{df}{dt} = -10t^2 + 8t + 1$

□□□□□□□▶□

Question No: 8 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & If \ \ x \geq 0 \ and \ \ y \geq 0 \\ 1 & Otherwise \end{cases}$$

□□□□□□□▶□ $f(x, y)$ is continuous at origin

□□□□□□□▶□ $f(0, 0)$ is not defined

□□□□□□□▶□ $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist

□□□□□□□▶□ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 9 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

□□□□□□□▶□parallel

□□□□□□□▶□**perpendicular**

□□□□□□□▶□opposite direction

□□□□□□□▶□No relation between them.

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ▶ perpendicular
- ▶ parallel

► **in opposite direction**

Question No: 11 (Marks: 1) - Please choose one

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

► **Absolute maximum and absolute minimum value**

► Relative maximum and relative minimum value

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► **Relative maximum at** (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

- f has relative maximum at (x_0, y_0)
- f has relative minimum at (x_0, y_0)
- f has saddle point at (x_0, y_0)
- **No conclusion can be drawn.**

Question No: 14 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- $x \neq y$
- $x \leq y$
- $x > y$

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of ----------

- Curve
- **Surface**
- Sphere
- Cone

Question No: 16 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- $\iint_{R} f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iint_{R} f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

Question No: 17 (Marks: 1) - Please choose one

If $R = \{(x,y)/0 \le x \le 2 \, and \, 1 \le y \le 4\}, then$

$\iint_{R} (6x^2 + 4xy^3)dA =$

$\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dydx$

▶

$\int_{0}^{2}\int_{1}^{4}(6x^2+4xy^3)dxdy$

▶

$\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dxdy$

▶

$\int_{2}^{4}\int_{0}^{1}(6x^2+4xy^3)dxdy$

▶

Question No: 18 (Marks: 1) - Please choose one

If $R=\{(x,y)/0\le x\le 2 \text{ and } -1\le y\le 1\}$, *then*

$\iint_{R}(x+2y^2)dA=$

$\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$

▶

$\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$

▶

$\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$

▶

$$\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$$

▶

Question No: 19 (Marks: 1) - Please choose one

If $R=\{(x,y)/0\le x\le 2 \text{ and } 0\le y\le 3\}$, *then*

$$\iint_R (1-ye^{xy})dA=$$

$$\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$$

▶

$$\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$$

▶

$$\int_{2}^{3}\int_{0}^{0}(1-ye^{xy})dxdy$$

▶

$$\int_{0}^{2}\int_{2}^{3}(4xe^{2y})dydx$$

▶

Question No: 20 (Marks: 1) - Please choose one

If $R=\{(x,y)/0\le x\le 4 \text{ and } 0\le y\le 9\}$, *then*

$$\iint_R (3x-4x\sqrt{xy})dA=$$

$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$

▶

$\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$

▶

$\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$

▶

$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

▶

Question No: 21 (Marks: 2)

Evaluate the following double integral.

$\iint \left(2xy + y^3\right) dx\, dy$

Question No: 22 (Marks: 2)

Let $f(x, y) = 2 + x^2 + \frac{y^2}{4}$

Find the gradient of f

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$$\iint \left(3+2x-3y^2\right)\, dx\, dy$$

Question No: 24 (Marks: 3)

Let $f(x,\ y,\ z) = yz^3 - 2x^2$
Find the gradient of f.

Question No: 25 (Marks: 5)

Find Equation of a Tangent plane to the surface $f(x,\ y,\ z) = x^2 + 3y + z^3 - 9$ at the point $(2,\ -1,\ 2)$

Question No: 26 (Marks: 5)

Evaluate the iterated integral

$$\int_2^4 \int_{\frac{x}{2}}^{\sqrt{x}} \left(xy\right)\, dy\, dx$$